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Classification of flash crashes using the Hawkes(*p*,*q*) framework*

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We introduce a novel modeling framework—the Hawkes(p, q) process—which allows us to parsimoniously disentangle and quantify the time-varying share of high frequency financial price changes that are due to endogenous feedback processes and not exogenous impulses. We show how both flexible exogenous arrival intensities, as well as a time-dependent feedback parameter can be estimated in a structural manner using an Expectation Maximization algorithm. We use this approach to investigate potential characteristic signatures of anomalous market regimes in the vicinity of 'flash crashes'—events where prices exhibit highly irregular and cascading dynamics. Our study covers some of the most liquid electronic financial markets, in particular equity and bond futures, foreign exchange and cryptocurrencies. Systematically balancing the degrees of freedom of both exogenously driving processes and endogenous feedback variation using information criteria, we show that the dynamics around such events are not universal, highlighting the usefulness of our approach: (i) post-mortem, for developing remedies and better future processes—e.g. improving circuit breakers or latency floor designs—and potentially (ii) ex-ante, for short-term forecasts in the case of endogenously driven events. Finally, we test our proposed model against a process with refined treatment of exogenous clustering dynamics in the spirit of the recently proposed autoregressive moving-average (ARMA) point process.

Keywords: Flash crash; Hawkes process; ARMA point process; High frequency financial data; Market microstructure; EM algorithm; Time-varying parameters

JEL Classifications: C01, C40, C52

1. Introduction

On 6 May 2010, a then unique intraday price drop in U.S. equity markets to lows of 9–10% below the market's opening price within a timespan of just over four minutes shook financial markets. The event was accompanied by significant spikes in trading activity in US equity, equity derivatives and equity futures markets. In fact, it was later concluded that the crash could be attributed to a large, aggressive sell order executed in the E-mini S&P futures market, which upset liquidity conditions to the point of cascading into so-called 'hot-potato trading'† (Commodity Futures Trading Commission-Securities Exchange Commission (CFTC-SEC) 2010a, 2010b). The order flows at this stage apparently grew ever more toxic (Easley *et al.* 2011, Andersen and Bondarenko 2014), undermining the willingness of market makers to hold inventory (Kirilenko *et al.* 2017) and prices collapsed as a result. The event, post-mortem coined as a 'flash crash', would become the first of its kind, followed by (supposedly) similar events in other financial instruments since then. For example on 15 October 2014, the price

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† Referring to the repeated passing of inventory imbalances between market participants.

for U.S. Treasury futures contracts spiked sharply, resulting in the benchmark 10-year contract's yield experiencing a 16-basis-point drop and then rebound within only a few minutes (USDT, BGFRS, FRBNY, SEC and CFTC 2015). Also in over-the-counter traded financial instruments, like foreign exchange, such events have occurred (Bank for International Settlements 2017a), making clear that they are a feature of modern electronic financial markets, independent of the underlying asset. These large, precipitous and transitory price drops generated significant concern among legislators, regulatory bodies and consequently spurred interest in the phenomenon by both the investing public and academia (Madhavan 2012, Cespa and Foucault 2014, McInish *et al.* 2014, Kyle and Obizhaeva 2019).

A common culprit for the incidence of flash crashes—in particular also in the investing public† —is HFT activity (Bouveret *et al.* 2015). While there is however little empirical evidence that these market participants significantly alter their inventory dynamics when faced with large liquidity imbalances, they are however found to amplify volatility in such situations (Kirilenko *et al.* 2017, Schroeder *et al.* 2020). Other aspects that are cited to encompass, amplify or even cause flash crashes are procyclical trading strategies (von der Becke and Sornette 2011), the exhaustion of risk constraints and an evaporation of market depth (Anderson *et al.* 2015, Bank for International Settlements 2017b), a breakdown of cross-market arbitrage activity (Menkveld and Yueshen 2019), data feed anomalies (Aldrich *et al.* 2017) and other mechanistic amplifiers like circuit breakers in related markets or hedging flows (Bank for International Settlements 2017a). Clearly, all of the aforementioned mechanisms can lead to a sudden deterioration of liquidity.‡ Furthermore, any significant shock to liquidity has the potential to become self-reinforcing. For example, as liquidity dries up, the expectations of market makers may change to the point where finding trading partners to clear their exposures may seem increasingly unlikely, which will cause them to withdraw or extremely skew their prices, thus further exacerbating the liquidity crisis and potentially destabilizing the price formation process to the point of breakdown. Such endogenous emergence of market crashes§ through imbalances between liquidity demand and risk-bearing capacities of market makers is also theoretically well understood (Huang and Wang 2009).

The picture that emerges is that the dynamics during such an event are resulting from a highly complex interaction between the behaviors of different market participants and a growing number of technical factors—owing to an ever-increasing technological sophistication and structural fragmentation of financial markets—each of which could have unforeseeably precipitated or amplified a critical event and a cascade of price changes. The involved feedback loops can in turn materialize in highly endogenous price process dynamics where, when pushed to a critical point, rapid movements from the minimum tick size all the way to fluctuations of the order of unity in the price are possible. Due to the increased connectedness between marketplaces, such extreme events are also more likely to be of systemic nature (Calcagnile *et al.* 2018). In this spirit, this paper refrains from trying to identify potential causes of flash crashes (in fact we hypothesize, and evidence suggests, that the possible triggers for such extreme price dynamics are manifold), but attempts to characterize the activity dynamics around such events in detail. In particular, we are interested in investigating potential instantaneous characteristic signatures of anomalous market regimes, which could foreshadow significant short-term market instabilities.

Clearly, a very natural and rich description of the computerized and fragmented markets under consideration here could be gained by adopting a complex systems perspective and employing some sort of agent-based model (Samanidou *et al.* 2007, Huang 2015, Dieci and He 2018). Indeed, agent-based models have been found to reproduce a range of phenomena that are associated with flash crashes. In minority games for example, the crowding phase provides an example of how hot-potato trading emerges as a result of the dependence between recent individual actions and aggregate behavior (Coolen 2005, Challet *et al.* 2013). Also, an amplification of volatility through adapting behavior of agents has been successfully documented in an agent-based setup (Harras and Sornette 2011). The approach we propose in the sequel can be viewed as being between an agent-based model and a purely stochastic model, as it captures the collective activity process resulting from the interactions of all agents. In this representation, each event caused by one market participant is a new signal to other market participants, which react themselves in a causal manner. The time-dependence of the relevant feedback parameter of the model can hereby be interpreted as the varying nature of the extent of that response—either due to changes in the behavior of agents or their composition in the market. Our framework is an extension of the classical Hawkes process (Hawkes 1971b, 1971a), which has proven to be a very useful framework to statistically measure and describe the interaction between events at the lowest time scale in a variety of scientific fields. In particular, this self-excited point process—where applicable—allows one to quantify and disentangle the relative contributions of endogenous (endo) and exogenous (exo) activities within a given system. Thanks to its mapping to a branching process (Ogata 1981), the classical Hawkes model provides a single parameter—the branching ratio η—which directly measures the degree of endogenous feedback effects in the arrival of events. In the context of fluctuations in high frequency financial prices, Filimonov and Sornette (2012) have opened the road toward the full utilization of the dynamics of the branching ratio in order to diagnose different regimes and to possibly forecast impending crises associated with the approach to criticality, which is determined by $\eta = 1$. Here, we leverage recent methodological developments in Wheatley *et al.* (2019) and Wehrli *et al.* (2020) and extend the framework proposed

† A survey by BlackRock (2010) for example reports that 46% of surveyed professionals believed high frequency trading (HFT) was the biggest contributor to the E-mini crash.
‡ We use the term liquidity quite casually here, without a strict definition and simply aim to describe an 'ease of trading'.
§ We will use the term crash generically for the resulting price dynamics in such events. While most price moves during such events indeed were negative—mostly by construction, as the asset under consideration is traded against cash—there is however no reason why upward price moves should not be a consequence of such a deterioration in liquidity. And in fact, in our empirical study, we will also look at examples where the considered price spiked up.

therein to allow estimation of the full dynamics of $\eta(t)$ in a structural manner.

The study of feedback parameters, such as the branching ratio, in order to characterize interactions and the stability of systems, in the same spirit as we do here, has a rich history in many scientific fields. Sornette (2006) provides a study of general precursory and recovery signatures accompanying endogenous and exogenous shocks in complex networks. Theoretical predictions for signatures of precursors and responses to shocks in very general processes were developed in Sornette and Helmstetter (2003), also providing a mean-field theory for a variant of the stochastic process we will use for the purposes of this study. A key insight is hereby that responses to shocks depend on the nature and extent of the feedback mechanisms present in the system dynamics. In Crane and Sornette (2008) for example, these differences are leveraged to classify dynamics in social systems as endogenous or exogenous. Other applications encompass book sales (Deschatres and Sornette 2005), seismicity (Helmstetter *et al.* 2003), or financial crashes (Sornette 2017). Naturally, the study of financial volatility time series also lends itself to such considerations (Sornette *et al.* 2004b, Joulin *et al.* 2008). For example in the recently proposed jump-diffusion model of Bates (2019), large market movements are the result of the accumulation of rapid and self-exciting intraday increases in conditional volatility, a quantity closely related to the conditional intensity of price changes used to describe the price process in the sequel of this paper. A recent study in Fosset *et al.* (2020) further links the occurrence of liquidity crises to such endogenous feedback effects and finds that, for these events to occur, financial markets must either constantly operate just below criticality, or exhibit time-dependent endogenous dynamics that occasionally visit the critical regime in order to produce the events in question. Resonating with the latter scenario and tying back to the argument of approximating aggregate agent behavior using Hawkes processes, also in agent-based models, the parameter controlling the social coupling/herding strength of traders must be allowed to intermittently reach the critical regime in order to reproduce stylized facts of financial markets like volatility clustering, realistic bubbles and corrections (Kaizoji *et al.* 2015, Westphal and Sornette 2020). In this spirit, our present study in fact proposes a framework to empirically distinguish a wide range of such dynamics on a case-by-case basis.

To test our approach as broadly as possible, we investigate events in equity and bond futures, as well as in the interbank foreign exchange market, all of which are among the most liquid financial markets traded electronically today. In particular, we investigate the E-mini flash crash of 6 May 2010, the U.S. Treasury yield flash crash of 15 October 2014 and several crashes in interbank currency spot markets. To complement our study of flash crashes in these markets, we also look at an event in the Ethereum cryptocurrency market. Since their creation in 2008, cryptocurrencies have significantly gained in attention and trading activity, with in excess of 100 cryptocurrencies currently traded in the market. In particular, Bitcoin and Ethereum (ETH) have assumed a leading role, making up more than half of the market capitalization of the top 1000 cryptocurrencies, even after the drastic decline in Bitcoin's market share in early 2017 (Gerlach *et al.* 2019). We propose the study of a flash crash in one of these markets to provide an important complement to the study of traditional financial assets for two reasons. First, the notion, that crashes appear to occur also in the absence of any particular exogenous event, suggests that—as discussed—destabilising feedback loops of behavioral origin may be present. The lack of a way to reliably assess the fundamental value of a cryptocurrency, as opposed to the other asset classes considered here, makes such a behavioral root of crashes a plausible hypothesis. Second, while equities, bonds and foreign exchange are highly mature markets, cryptocurrencies have only been around for a good decade, and their marketplaces have a much more contemporary technological foundation. As such, any commonalities arising from the joint study of mature and adolescent markets will point towards mechanisms that are indeed more fundamental than simple differences in asset characteristics or trading technologies. Furthermore, to our knowledge, this is also the first study to provide a comprehensive cross-asset analysis of flash crash dynamics and their exogenous and endogenous constituents. Finally, it is important to note that we select events without a systematic definition of what exactly constitutes a flash crash and simply look at events that were designated as such by the investing public and the financial press.

The rest of the paper is organized as follows. Section 2 describes the data sets used in the study and provides some institutional details of the various financial markets investigated here. Section 3 introduces the point process model used to characterize price formation dynamics and section 4 summarizes the methodology—described in detail in a technical appendix—used to estimate this model on high frequency price data. Section 5 surveys a series of flash crashes and provides estimated dynamics of exogenous and endogenous market activity around the events. Section 6 explores the goodness of the fitted processes and tests the proposed model against an alternative specification with a refined treatment of exogenous clustering. Section 7 concludes.

2. Data sets

Since the financial instruments we are studying might be traded in different places/on different platforms simultaneou sly—foreign exchange for example being an OTC market with a highly fragmented market structure (Bank for International Settlements 2017, Schrimpf and Shushko 2019)—we aim to base our analysis on data from the most liquid marketplace available for each instrument. In the sequel, we provide a description of the data sets we rely on—with a particular focus on technical aspects that could influence observed activity patterns—and discuss some institutional details of the various markets.

2.1. *The 10-year U.S. Treasury note futures contract*

Treasury futures are standardized contracts to trade U.S. government debt obligations for future delivery or settlement. They were introduced in the 1970s at the Chicago Board of Trade (CBOT), now part of the Chicago Mercantile Exchange

(CME), as a tool to hedge short-term risks on U.S. treasury yields. One contract for the 10-year futures contract has a face value at maturity of USD 100 000. That is, each contract affords the buyer the right to buy an underlying Treasury note with a face value of USD 100 000. The prices of these contracts are quoted as percentages of their par value, in the case of the 10-year contract the minimum price increment is 1/64%. In USD terms, this amounts to USD 15.625. Treasury futures are traded on an electronic limit orderbook, the CME Globex trading platform, but there are also still open outcry trading pits, which we however exclude from the analysis as they only account for a negligible share of trading. Overall, the Treasury futures market is a highly computerized financial market, where high frequency trading is estimated to account for well more than half of the trading activity on benchmark contracts (Bouveret *et al.* 2015). Trading on the Globex platform happens continuously from 5 pm (Chicago Time; CT) to 4 pm (CT) of the next day, Sunday to Friday. Between 4 pm and 5 pm CT, the exchange imposes a maintenance window where no trading is possible. Treasury futures contracts have a quarterly expiration cycle in March, June, September and December. At any given point in time, several contracts written on the same underlying security (differing only in their expiration dates) may trade side by side. The contract closest to expiry, the so-called front-month contract, offers the most liquidity and is the contract we focus our analysis on. In October 2014, when the flash crash occurred, the front-month contract was the one expiring in December (symbol ZNZ4). In principle, the data feed also contains quotes from a so-called implied orderbook, which originate from some order book related to the futures contract under consideration. For our study, we ignore quotes from the implied order book and compute mid-prices from the direct order book exclusively. Our data set is obtained from the CME DataMine and contains all market data messages required to reconstruct the limit order book tick-by-tick, 10 price levels deep. Each market data message is timestamped to the millisecond at the time of market data publication, but can contain multiple modifications to the book. Each modification is additionally given a timestamp when the respective change to the book actually occurred on the Globex exchange, but only timestamped to the second.

2.2. *The E-mini S&P500 futures contract*

The E-mini S&P 500 futures contract was introduced in 1997 by the CME as a complement to the regular S&P 500 futures contract and has become one of the most actively traded securities in the world. It trades exclusively electronically on the CME Globex platform. As for U.S. Treasury futures, the share of high frequency trading is typically estimated to be 50% or higher (Sussman *et al.* 2009, Aite Group, New World Order 2009)—even though there are some issues surrounding estimates of the size of high frequency trading volumes (SEC 2010). Trading is possible from 5 pm CT to 4 pm CT of the next day, Sunday to Friday. In addition to the one hour maintenance period starting at 4 pm, there is an additional temporary trading halt of 15 minutes, starting at 3:15 pm CT. The size of S&P 500 E-mini contracts is one-fifth of the regular S&P 500 futures contracts—which are still partially floor-traded—and it has a notional value of USD 50 times the value of the S&P 500 stock index, denominated in index points. The contract has a tick size of 0.25 index points (USD 12.50) and four expiration months per year. We use data for the front-month contract to ensure that we use the most actively traded contract for our analysis, which in May 2010 was the contract with expiry in June. Our data set for the E-mini is obtained from TickData,† which offers research-quality, historical tick-by-tick prices of futures and index markets. The data set contains entries for the best bid and ask quotes whenever either the respective price or amount changed, timestamped with millisecond resolution.

2.3. *Foreign exchange spot interbank market*

For the events in foreign exchange markets, we rely on data from the trading venue which is considered as the respective primary interbank trading place for the currency pair. In particular, this is the Electronic Broking Services (EBS) interbank trading platform for the EUR/USD and USD/JPY currency pairs and Thomson Reuters Matching, now a service of Refinitiv, for the other currency pairs in the present study. All currency pairs analyzed are also traded on a plethora of other trading venues simultaneously, but EBS and Refinitiv are still considered primary marketplaces and important points of price discovery, despite their declining market share (Bank for International Settlements 2017). The EBS dataset provides a snapshot of the limit order book, whenever a change occurs in the 10 best price levels, within the last 100 milliseconds, with corresponding timestamp resolution of 100 milliseconds. The data for the Refinitiv currency pairs is obtained from Thomson Reuters Tick History (TRTH). An entry encompassing the best quoted price levels on each side of the market is available when either the best available price or respective quoted amount has changed. The data set contains two timestamps for each modification, one representing the time when the change happened on the trading platform with millisecond resolution and one taken at collection time by TRTH with nanosecond resolution. Both foreign exchange trading platforms operate continuously from Sunday to Friday and their minimum order sizes are one million of the base currency. Prices on EBS are quoted in half-pips,‡ whereas on Reuters the tick size is typically one pip.§

2.4. *The Ethereum cryptocurrency market*

For the study of the Ethereum cryptocurrency—or rather the Ethereum platform's native cryptocurrency 'Ether'—we look at data from GDAX (Global Digital Asset Exchange), the first licensed U.S. bitcoin exchange and largest market to trade Ether against the US-Dollar in the world, launched in 2016. Even though technically speaking, the Ethereum platform itself is not a cryptocurrency, we will use the names Ethereum

† www.tickdata.com
‡ One pip being 0.0001 USD for EUR/USD and 0.01 JPY for USD/JPY.
§ Being 0.0001 units of the price currency for the considered currency pairs, except for USD/ZAR, where the tick size is 0.0005 South African Rand.

and Ether interchangeably. We obtain data on quote and trade activity on the GDAX order book from coinapi.io, a provider of unified data APIs to cryptocurrency markets. The data set contains tick-by-tick information on the top-of-book quoted prices and amounts, as well as all trading activity on the GDAX exchange. For each update, two timestamps are available, one provided by the GDAX exchange with millisecond resolution and one with even higher resolution, taken when the market data update was recorded by the data provider. The ETH/USD pair is traded with price increments of 0.01 USD. As opposed to the other financial markets considered here, ETH/USD can be traded on the GDAX 24/7.

2.5. Sample selection and summary

From each of the data sets, we compute mid-prices, i.e. the average of the best prevailing bid and ask price at any given point in time. To fit the mid-price data for the various events, we collect the times where the mid-price changed, in a window of one day centered around the trough/peak of each event. Since not all of our data sets encompass 12 hours of pre- and post-event data—or include a weekend/non-trading period within the desired window—not all samples will be 24 hours long. We provide a short summary of the events under consideration and the respective sample sizes in table 1. Additionally, we illustrate the distribution of the events across the day in figure 1, which shows that, while there are indeed some events that happen during traditionally illiquid times of the day, there are also crashes that occurred during the more liquid times when trading centres in both Europe and the United States are active.

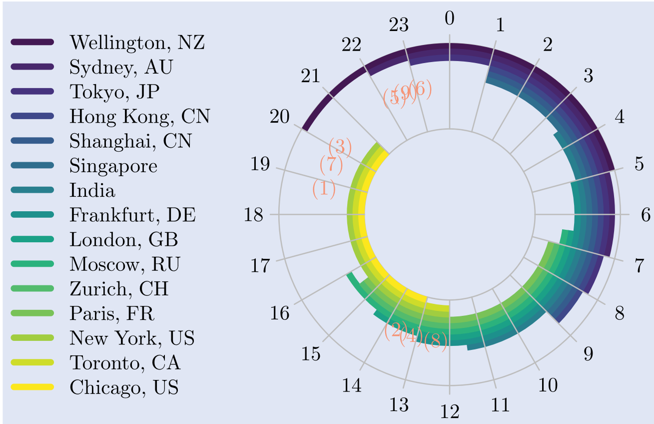


Figure 1. Shows a Coordinated Universal Time (UTC) clock of the trading hours of major financial centers, where we have used the representation of Chen and Stindl (2018). The events from table 1 are indicated with their number at the respective time of day.

3. The Hawkes model with time-dependent endogenous dynamics

The classical Hawkes process (Hawkes 1971a, 1971b) has provided an important tool in the study of financial price fluctuations (see e.g. Bowsher 2007, Bacry *et al.* 2013, Bacry and Muzy 2014, Rambaldi *et al.* 2019 and reviews in Bacry *et al.* 2015 and Hawkes 2018). In particular, it parsimoniously allows one to distinguish endogenous responses from exogenous influences (Filimonov and Sornette 2012, Hardiman *et al.* 2013, Filimonov *et al.* 2014, Hardiman and Bouchaud 2014, Filimonov and Sornette 2015b)—a task that we call 'solving the endo-exo problem' (Wheatley *et al.* 2019, Wehrli *et al.* 2020). In the standard Hawkes model, the parameter η—being defined most naturally in the language of branching processes (Harris 1963, Daley and Vere-Jones 2008) as the average number of offspring of first generation triggered per observed point—characterizes also the degree of self-excitation inherent to the dynamics of the system. At the same time, η characterizes the fraction of triggered events in the observed population (Helmstetter and Sornette 2003, Saichev and Sornette 2006). When $\eta < 1$, then the process is *subcritical* and, starting from a single immigrant point at some time, dies out with probability one. When the process is *super-critical* ($\eta > 1$), the process has finite probability to explode to an infinite number of points (Saichev *et al.* 2005). The critical regime ($\eta = 1$) delineates the transition between the two other regimes and can still produce stationary dynamics under some conditions for the intensity of exogenously occurring events (Brémaud and Massoulié 2001, Saichev and Sornette 2014, Jaisson and Rosenbaum 2015).

Table 1. Overview of data sets used in the study.

#	Market	Date	T	N_T	$\langle\lambda\rangle$	$\bar{\lambda}$
(1)	E-mini	6 May 2010	13 h 30 min	86 486	1.78	172
(2)	U.S. Treasury	15 October 2014	19 h 20 min	108 833	1.56	270
(3)	EUR/USD	18 March 2015	24 h	68 832	0.80	10
(4)	NZD/USD	24 August 2015	24 h	12 861	0.15	2
(5)	USD/ZAR	11 January 2016	16 h 18 min	5427	0.09	2
(6)	GBP/USD	7 October 2016	24 h	27 169	0.31	4
(7)	ETH/USD	21 June 2017	12 h 23 min	110 856	2.49	31
(8)	EUR/USD	25 December 2017	24 h	869	0.01	5
(9)	USD/JPY	2 January 2019	24 h	44 782	0.52	10

Notes: The fitting window T of each event is the result of taking one day of data around the trough/peak of the event. Where the sample duration is less than one day, this means that either: the resulting sample window covers a weekend/non-trading period; or is limited by the size of the available data set. The number of events within each window is denoted N_T. We provide average unconditional event intensities $\langle\lambda\rangle$, as well as the respective maximum $\bar{\lambda}$ in the sample. The units of the intensities are sec^{-1}.

Here, we consider a generalization of the standard Hawkes process, which features both exogenous non-stationarity (in the form of a time-dependent, deterministic arrival intensity for new exogenous points) as well as time-dependent endogenous dynamics in the form of a deterministically varying branching ratio. The point process N we work with is defined as a random measure on $\mathbb{R}$ and counts the random number of points in some Borel set $A \subseteq \mathbb{R}$, denoted $N(A)$. For ease of notation, we write $N_t := N((0,t])$ for $t \in [0,\infty)$. A single realization of the process consists of all the times $s \in [0,t]$, where $\mathrm{d}N_s = N_s - \lim_{u\to s^-} N_u = 1$ and we assume that the process is orderly. The history of the process is given by $\sigma(N) = (\mathcal{F}_t)_{t\in(-\infty,\infty)}$, where $\mathcal{F}_t$ is the σ-algebra generated by N up to and including t, i.e. $\mathcal{F}_t = \sigma(N_s, s \le t)$. The $\mathcal{F}$-conditional intensity function, which fully defines our process, has the form

$$\lambda(t) = \mu(t) + \int_{-\infty}^{t} \eta(s)h(t-s)\mathrm{d}N_s$$
$$= \mu(t) + \sum_{i:t_i<t} \eta(t_i)h(t-t_i), \qquad (1)$$

where $\mu(t) \ge 0 \quad \forall t$ is the intensity of exogenously arriving points and can be called the immigration intensity in view of the branching process interpretation of the Hawkes process (Ogata 1981). The endogenous feedback effects of the process are defined by the instantaneous branching ratio $\eta(t) \ge 0 \quad \forall t$ and an offspring density $h : [0,\infty) \to [0,\infty)$, which has an L^1 norm of one. It is worth considering some possible specifications for the instantaneous branching ratio, in order to delineate properly the class of models we are considering here.

(i) A constant function $\eta(t) \equiv \eta$ recovers the classical Hawkes process.
(ii) If $\eta(t)$ is some fixed, deterministic function of time, then there can be time-varying trends and shifts in the expected number of first generation offspring that are triggered by a given point.
(iii) An often considered general specification defines $\{\eta(t_i)\}_{i\ge1}$ to be a sequence of independent and identically distributed random variables, e.g. also encompassing the dynamic contagion process of Dassios and Zhao (2011).
(iv) If $\eta(t)$ is allowed to evolve according to a stochastic differential equation, then we recover the Hawkes process with stochastic excitations from Lee *et al.* (2016).
(v) The instantaneous branching ratio can also be envisaged to depend on some other observed quantity beyond their occurrence time, which would cast the setting into a marked point process specification (Daley and Vere-Jones 2003). The marks m_i associated with some point t_i relate to the instantaneous branching ratio through a ‘productivity law’, given by some function f, as $\eta(t_i) = f(m_i)$. This specification for example includes the well-known epidemic-type aftershock (ETAS) model of Ogata (1988) developed in statistical seismology to describe complex sequences of earthquake aftershocks, where the productivity is an exponential function of the magnitude m_i of the main shock.
(vi) The specification $\eta(t) = f(\lambda(t))$, where the function $f(\cdot)$ encodes the dependence of the branching ratio on the contemporaneous conditional intensity, corresponds to the recursive point process model of Schoenberg *et al.* (2017), where the branching ratio depends on the conditional intensity, which itself is a function of the instantaneous branching ratio of preceding points.
(vii) A renewal-type version of the instantaneous branching ratio can be obtained by choosing $\eta(t_i) = f(t_i - t_{i-1})$, e.g. applied in Harte (2014).

In our setup here, $\eta(t)$ is a *deterministic* function of time as in (ii), which is sampled at random jump times t_i. We will hereby model the function describing the time-evolution of η in a flexible, non-parametric fashion.

Also in this specific setup, we can interpret λ as a conditional hazard function, i.e. $\lambda(t) = \lim_{\Delta\to0} \Delta^{-1}\mathbb{E}[N_{t+\Delta} - N_t|\mathcal{F}_{t-}]$. The branching representation of this point process is furthermore very similar to the standard Hawkes process in that $\mu(t)$ induces immigrants according to an inhomogeneous Poisson process. All existing points t_i then trigger a first generation of offspring with intensity $\eta(t_i)h(t-t_i)$, which in turn trigger their own generation of offspring in the same manner, and so on. The collection of these independent Poisson processes forms the process N and the $\eta(t)$ are the expected number of immediate offspring of a single point occurring at time t. This branching representation also provides the basis for efficient simulation† of the process. Our model for the endogenous feedback mechanism—i.e. the memory of the process—is an approximate Pareto distribution (Hardiman *et al.* 2013) with tail index $\varepsilon > 0$,

$$h(t) = Z^{-1}\left[\sum_{i=0}^{M-1} \xi_i^{-(1+\varepsilon)} e^{-\frac{t}{\xi_i}} - S e^{-\frac{t}{\xi_{-1}}}\right]. \qquad (2)$$

The $\xi_i = \tau_0 b^i$ are power law weights, $\tau_0 > 0$ a short-lag cut-off and S and Z normalizing factors. The range over which the power law is approximated can be controlled by M and b. If one is, e.g. interested in computing the log-likelihood of the process (Daley and Vere-Jones 2008) for an observed sequence of events at times $t_i \in (0,T]$,

$$\log L(t_1,\ldots,t_{N_T}) = \sum_{i=1}^{N_T} \log\lambda(t_i) - \int_0^T \lambda(t)\mathrm{d}t, \qquad (3)$$

then this approximate power law form for the offspring density allows a reduction of computational complexity from $\mathcal{O}(N^2)$ to $\mathcal{O}(N)$ (Ozaki 1979).

† The process can be simulated recursively, generation-by-generation. The immigrant points from $\mu(t)$ are simulated first—either by thinning Lewis and Shedler (1979) or by drawing the number of points on $[0,T]$ from the Poisson distribution with mean $\int_0^T \mu(t)\mathrm{d}t$ and choosing their time-location by inverse-transform sampling (Cinlar 2013) from a density $\propto \mu(t)$. The number of points for each offspring process of a given mother t_i are then each drawn from a Poisson distribution with mean $\eta(t_i)$ and dispersed on the time axis according to random inter-event times drawn from $h(t)$.

4. Calibration to mid-price data—the Hawkes(p, q) model

Our goal is to estimate the model (1) based on realizations $\{t_i\}_{i=1}^{N_T}$ of the random sequence of event times induced by the process of high frequency mid-price changes. The predominant approach to create such time-varying estimates is to numerically maximize the log-likelihood (3) of a time-invariant parameter model on windows of some fixed length (i.e. on sub-samples of the jump times t_i) and rolling them forward in time to trace out a time-variant estimate of the parameters (e.g. Filimonov and Sornette 2012, Hardiman *et al.* 2013, Filimonov *et al.* 2014, Mark *et al.* 2020). This approach of rectangular windowing brings several practical difficulties, like the selection of window size in a way that does not neglect relevant memory, or treatment of edge and truncation effects, which when neglected can bias parameter estimates. Similar issues are for example well-known in the signal processing literature, e.g. in the form of the 'Gibbs phenomenon'. Resembling approaches in signal processing to cope with these issues, also for models such as (1), the rectangular windowing can be replaced by more general approaches like *local maximum likelihood* (Tibshirani and Hastie 1987, Fan *et al.* 1998), which introduces kernel-weights into the log-likelihood in order to localize its parameters. However, bandwidth selection for the kernel is a highly involved procedure when done systematically. Furthermore, the optimal bandwidth in general also depends on bias and variance estimates of the non-parametric estimator, which might not be easily available. Additionally, independent control over the flexibility of the various parameters in the model is also non-trivial.

For the present application, where adaptive and time-dependent estimators for both $\mu(t)$ and $\eta(t)$ are required, we prefer an approach where the flexibility of the individual models can be controlled and traded off more explicitly. Our approach is to rely on the Expectation Maximizaton (EM) algorithm (Dempster *et al.* 1977) to provide a maximum likelihood estimate (MLE). The EM algorithm is a widely applicable computational approach for the iterative construction of MLE in incomplete data or latent variable problems. In the context of point processes with affine conditional intensities like (1), the unobserved information can be chosen to be the corresponding branching structure of the point process. In this case, the EM decomposes the estimation into sub-problems, where at each iteration of the algorithm, separate models can be fitted for $\mu(t)$ and $\eta(t)$. Here, we will use adaptive spline models for both functions, which allow one to control their flexibility by choosing a number of degrees of freedom (i.e. knots of the splines). We denote these degrees of freedom as p for $\mu(t)$, and q for $\eta(t)$, labeling the resulting full model as a *Hawkes*(p, q) model. To perform model selection, one can then simply apply the estimation procedure for a range of values for p and q and take the model which minimizes some information criterion. Based on past results (Wheatley *et al.* 2019), we choose the Bayesian Information Criterion (BIC) (Schwarz 1978) for our purposes.

A detailed description of the EM estimation procedure is given in appendix 1. The ability of the approach to estimate the dynamics of interest in the presence of time-varying exogenous and endogenous excitation—in particular during a regime where the process approaches criticality temporarily—is furthermore demonstrated in appendix 2.

5. Endogeneity of financial markets around flash crashes

To characterize exogenous and endogenous dynamics around flash crashes, we now apply the Hawkes(p, q) methodology to our data sets from section 2. In the sequel, we provide a short description of each event—in chronological order—and discuss how endogenous and exogenous activity evolved through the lens of our Hawkes(p, q) fits. We begin by a short discussion on the treatment of timestamps used for estimation and provide the specific settings of the EM algorithm used in estimation.

5.1. *Preprocessing of timestamps and calibration specifics*

As was discussed in section 2, the data sets in our study exhibit different limitations regarding the resolution of the timestamps associated with the mid-prices. This brings several practical difficulties. First, any limitation in time resolution introduces a spurious discretization of the duration data, which might affect estimation of the offspring probability density during the M-step of the EM algorithm. Second, while for our foreign exchange and cryptocurrency data sets, even where timestamps are available with millisecond precision, the timestamps are in fact unique, for the E-mini and U.S. Treasury data set, around 45% of the events have identical timestamps. For our point process model to be well-defined, we need to assume that the associated counting process is *orderly* —i.e. $\mathbb{P}[N_{t+\Delta} - N_t > 1|\mathcal{F}_{t-}] = o(\Delta)$. As such, contemporaneously occurring events are incompatible with the definition of the process. The principled approach to deal with such limited timestamp resolution and/or duplicate timestamps is to randomize them within the interval of uncertainty (Filimonov and Sornette 2012, Hardiman *et al.* 2013, Lallouache and Challet 2016). Additionally, depending on the timestamp which is used as a base for the calibration, an additional source of uncertainty is introduced by a potential bundling of updates to the limit order book during data collection/processing, which when combined with inadequate assumptions on the related timestamp uncertainty can induce spurious clustering in the data (Filimonov *et al.* 2014, Filimonov and Sornette 2015b).

Here, we follow along the lines suggested in Filimonov and Sornette (2015b) and whenever available, consider timestamps provided by the exchange as a reliable source of data, despite their typically inferior resolution as compared to potential additional timestamps taken by the provider of the data sets. For all data sets except the E-mini, we have the exchange timestamps available and simply smooth the resulting duration data by randomizing their timestamps within the resolution of the respective timestamps, which is—as elaborated in section 2—$\Delta = 1$ second for the U.S. Treasury futures data, $\Delta = 100$ milliseconds for the EBS data and $\Delta = 1$ millisecond for the TRTH and Ethereum data. Additionally, for these data sets, we will also produce fits on the

Table 2. Overview of timestamp sources (EXCH = exchange, DP = data provider) and durations per event.

#	Market	Source	Median duration (sec)	Mean duration (sec)
(1)	E-mini	DP	0.043	1.090
(2)	U.S. Treasury	EXCH	0.006	1.128
(3)	EUR/USD	EXCH	6.7	99.480
(4)	NZD/USD	EXCH	2	6.716
(5)	USD/ZAR	EXCH	1	10.701
(6)	GBP/USD	EXCH	0.751	3.180
(7)	ETH/USD	EXCH	0.101	0.402
(8)	EUR/USD	EXCH	0.3	1.255
(9)	USD/JPY	EXCH	0.5	1.930

Note: For the U.S. Treasury data, the timestamps are chosen to be the ones taken at the time of publication of the market data.

durations resulting from raw timestamps (denoted $\Delta = 0$). For the U.S. Treasury futures data, we will use for this purpose the higher resolution timestamps taken at the time the market data update was published, but remove all duplicates. Additionally, we can look at the difference between the time a market data message was published (with millisecond precision) and the timestamp of each modification at the exchange contained within such a message (with second precision). This difference Δ_i might in fact give a more narrow definition of the uncertainty around the true timestamp of a mid-price jump than the one-second window assumed in the first set of fits. It turns out that the distribution of these time differences follows a uniform distribution quite well. As such, we produce an additional set of fits for the U.S. Treasury futures event, applying a random time shift $\Delta \sim U(0, \Delta_i)$ to each second-precision timestamp t_i, contained within a market data message used to derive Δ_i. For the E-mini data, we do not have any exchange timestamps available and thus rely on the timestamps with millisecond precision from the data provider, but perform a search over different assumptions on Δ. As suggested in Filimonov *et al.* (2014), when relying on such enriched timestamps, the choice of Δ can be informed by the distribution of durations. As can be seen from table 2, the median duration of the E-mini data is approximately 40 milliseconds, while its average is around one second. This suggests that a reasonable value for Δ is of the order of 100–1000 milliseconds.† In order to be as close as possible to the analysis in Filimonov and Sornette (2012), we will focus on $\Delta = 1$ second, but also consider alternative assumptions of $\Delta = 0$ (fitting on unique timestamps only, without additional randomization), $\Delta = 1$ millisecond and $\Delta = 100$ milliseconds and assess the effect of a particular choice for Δ on the branching ratio and in particular the resulting goodness of the fit.

Finally, in our EM estimation, we vary the allowed degrees of freedom for the immigration and branching ratio functions $p, q \in [1, 25]$ and, as in our simulation study before, truncate the interaction length of $h(\cdot)$ at 1000 points. The algorithm is run for 500 iterations, or early stopped as soon as convergence is achieved. Our criterion for convergence will be that the supremum norm of the change in the parameter vector from one iteration to the next stays below a threshold of 10^{-4} for a couple of consecutive iterations.

5.2. *The E-mini S&P 500 futures crash in May 2010*

We begin our study by looking at the E-mini flash crash of 6 May 2010. In figure 2, we show the evolution of the mid-price of the futures contract and the surrounding activity in quoting and trading activity. The spike in quoting activity is indeed very impressive with over 4000 mid-price changes per minute at the peak. Regarding the particular sequence of events that encompassed the crash, we refrain from repeating them here, as they are covered in great detail in Commodity Futures Trading Commission-Securities Exchange Commission (CFTC-SEC) (2010a) and Commodity Futures Trading Commission-Securities Exchange Commission (CFTC-SEC) (2010b) and several other studies. We want to highlight however the many technical frictions that accompanied the event. For example the fact that there was a significant dislocation between ETF prices on the NYSE and the E-mini futures on the CME is well-known, as is the fact that there was a five second trading halt on the CME due to its stop logic functionality during the event. What is less well covered and was recently highlighted in great detail in Aldrich *et al.* (2017) is that there were multiple message-level issues on the data feeds just a few minutes before the trading halt was triggered on the CME, leading to compromised market data integrity and market participants' withdrawal of liquidity. We believe that such noise introduced into the data feeds indeed has the potential to amplify feedback effects, much like other mechanical amplifiers do.

Looking at the resulting estimates in figure 2, we see that the optimal Hawkes(19,7) fit using $\Delta = 1$ second recovers a branching ratio that is rather stable around 0.7 before the crash and then rises to criticality as the mid-price diverges. The dynamics of the branching ratio are in very good agreement with what was recovered in Filimonov and Sornette (2012), using the same randomization window, even though their peak branching ratio of around 0.95 is slightly lower than what we find here. A transient rise in the branching ratio is also found

† We remark that Filimonov *et al.* (2014) find consistent uncertainties (based on data from a different data provider) of the timestamps for the E-mini in the year 2010 and additionally show that constant branching ratio estimates for different assumptions in the range 100 ms $< \Delta <$ 1 second are practically indistinguishable, while Δ = 1 second results in slightly higher branching ratios, but similar dynamics.

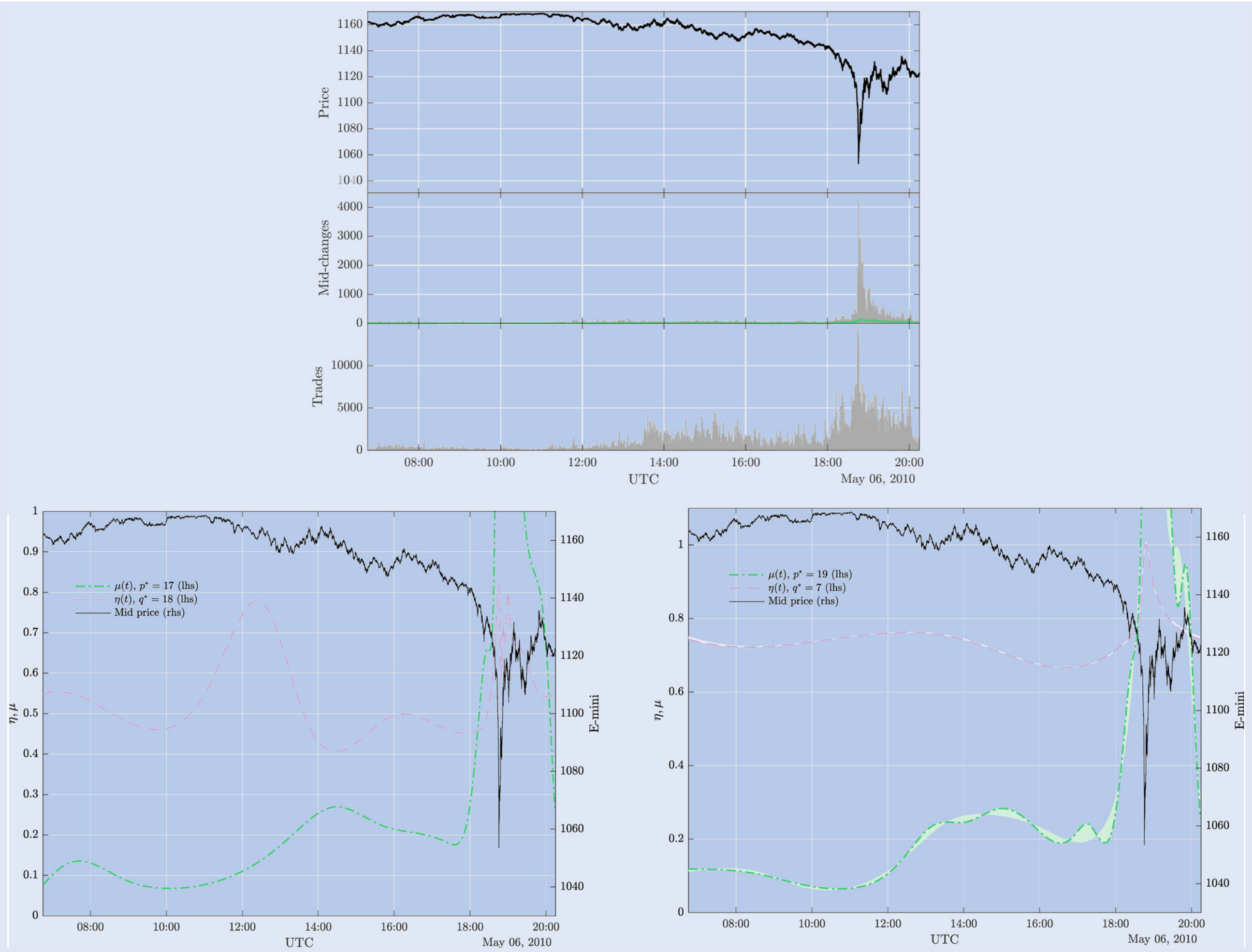


Figure 2. The top row shows the evolution of the mid-price of the S&P 500 E-mini futures contract around the event on 6 May 2010. Additionally shown below the price are the number of mid-price changes and trades per minute. In the bottom row, the estimated immigration intensities $\mu(t)$, in sec^{-1}, and branching ratio $\eta(t)$ are shown in green and purple for fits using $\Delta = 0$ (left) and $\Delta = 1$ second (right), respectively. The lines indicate the BIC-optimal functions and the shaded regions indicate 95% credible intervals, obtained by bootstrap sampling 1000 times according to the approximate posterior probabilities of the models, see Appendix 1. The BIC-optimal degrees of freedom for the immigration (p^*) and the branching ratio (q^*) are indicated in the legend. Furthermore, the optimal immigration estimate from fits using $\Delta = 1$ second is also indicated in green in the upper panel, scaled by integrating $\mu(t)$ within the respective bin of the mid-change counts.

when fitting on raw/unique timestamps ($\Delta = 0$). There, the optimal Hawkes(17,18) fit recovers a branching ratio that rises from values around 0.5 to ≈ 0.8 as the mid-price diverges. Unsurprisingly, removing almost half of the points to achieve uniqueness in the timestamps results in a downward bias of the average level of self-excitation of ≈ 0.2, but to a large extent preserves its dynamics. The branching ratio stops its rise and drops noticeably around the time the CME halted trading. Once trading resumed, the branching ratio relaxes towards pre-crash levels. The exogenous intensity on the other hand also explodes sharply as the crash materializes, but stays elevated for a much longer period of time than the branching ratio. Comparing the goodness of the fits for different assumptions on Δ (see also section 6) reveals that for all values except $\Delta = 0$, a significant amount of autocorrelation persists in the residuals of the process, even though the average estimates for $\eta(t)$ are comparable across the different assumptions and in line with other previous studies at around 0.7–0.8. Fits using $\Delta = 1$ second exhibit the least residual autocorrelation on randomized timestamps, thus produces the best goodness of fit among the candidates for Δ tested, and we use this value as the optimal choice for the purpose of this study.

5.3. *The U.S. Treasury bond yield crash in October 2014*

Between 13:33 and 13:45 UTC, on October 15, 2014, the U.S. Treasury bond market experienced a significant spike in prices, against the backdrop of falling yields that had already set in with the U.S. stock market opening one hour earlier. During the 12-minute period of the crash, the yield on the 10-year U.S. Treasury note traded in an extraordinary range of 37 basis points. Market activity was indeed extremely elevated at almost 3000 mid-price updates per minute at the peak (see figure 3). Despite these movements, there was no trading halt on the CME, bid-ask spreads remained fairly tight and prices showed very few gaps. USDT, BGFRS, FRBNY, SEC and CFTC (2015) highlights a number of structural factors relevant to the event, including the increasingly electronic nature of the U.S. treasury market and the prominent role

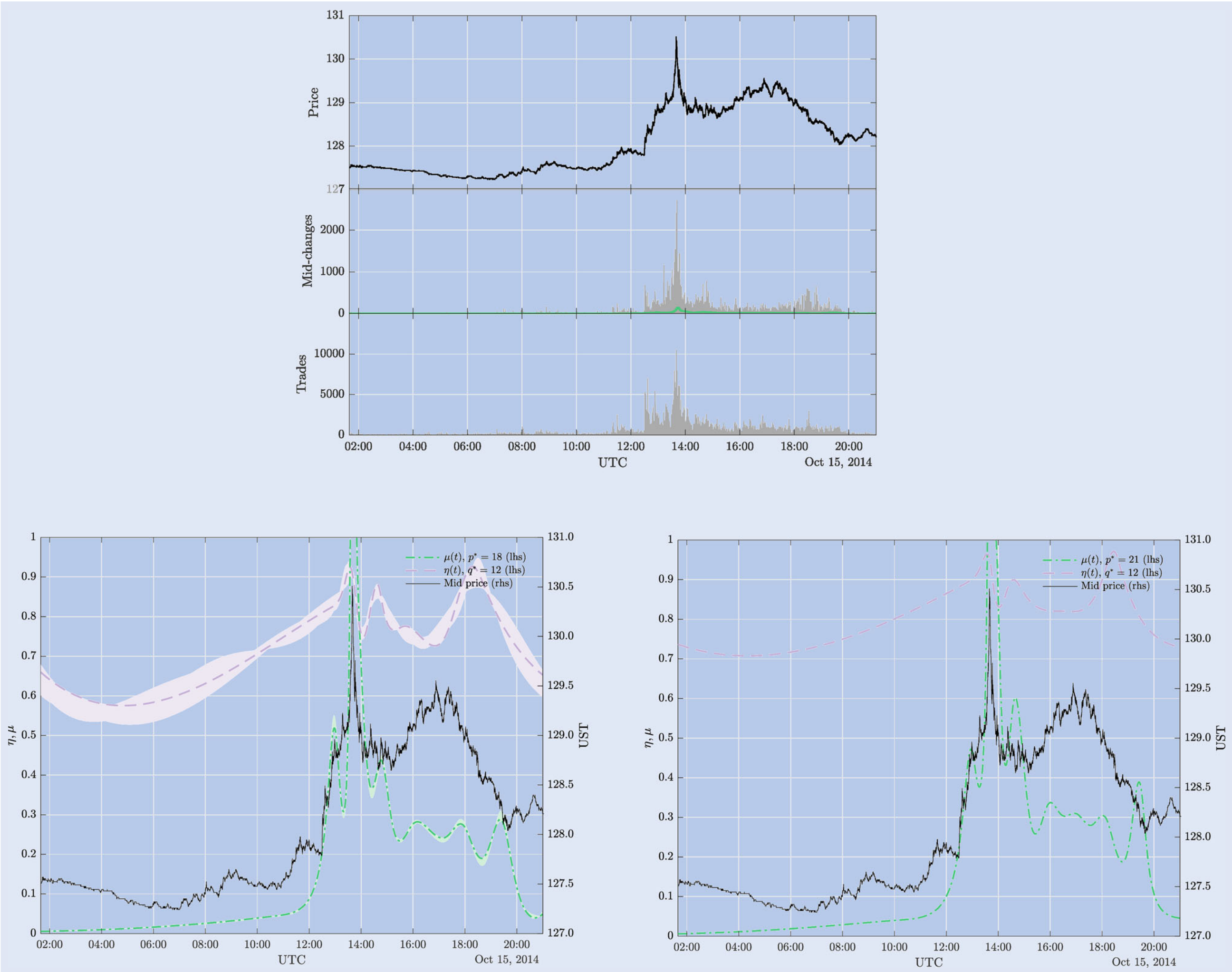


Figure 3. Same as figure 2 for the 10-year U.S. Treasury Note Futures contract event on 15 October 2014. The bottom panels show the estimated immigration intensities $\mu(t)$, in sec^{-1} and branching ratio $\eta(t)$, in green and purple, for fits using unique/raw timestamps ($\Delta = 0$; left) and using timestamps that are randomized assuming $\Delta = 1$ second (right), respectively.

of automated trading strategies. The report also documents that message traffic increased so dramatically just before the event that the average latency on the exchange built up from sub-milliseconds to around 60 milliseconds for a short time. Similar to the E-mini event, there was a substantial amount of hot-potato trading during the U.S. Treasury event. As such, many of the mechanisms that we suspect to result in highly endogenous price dynamics were also present during this event.

And indeed, the selected Hawkes(21,12) fit under the assumption of $\Delta = 1$ second identifies a branching ratio that peaks at ≈ 0.96 just as the mid-price also reaches its most extreme value. Assuming $\Delta \sim U(0, \Delta_i)$ recovers a Hawkes(19,12) model, which identifies a similar gradual rise of the branching ratio as a precursor to the crash, even rising up to $\eta(t) \approx 1$. Since, as for the E-mini, the goodness of fit under the assumption of $\Delta = 1$ second is superior, we will focus on that value in the rest of the paper. The endogenous dynamics we identify, irrespective of the assumption on Δ, calm down as the price recovers and then hours later in fact $\eta(t)$ rises again to near-critical levels without triggering a similarly drastic price response—the price in fact stabilizes further during this period. As we will see in other events subsequently, the branching ratio can indeed also rise in times when market dynamics recover from perturbations—a behavior that we think can be understood in the context of adaptive control dynamics as we will discuss below. The optimal Hawkes(18,12) on raw/unique timestamps ($\Delta = 0$) recovers very similar dynamics for the endogenous dynamics, with a slightly lower peak value of ≈ 0.93 for the branching ratio and some more uncertainty in model selection.

5.4. The US-Dollar crash in March 2015

Within minutes on 18 March 2015, the US-Dollar depreciated around 2% against the Euro and recovered most of this loss within a very short period of time thereafter. The depreciation of the US-Dollar had already started around two hours earlier after the Federal Open Market Committee (FOMC) had released a statement that was not aligned with market expectations. Simultaneous to the depreciation, U.S. stock markets traded sharply higher. The flash crash then occurred just a few minutes after the U.S. stock markets closed, visible at 20:00 UTC in figure 4. The event was accompanied

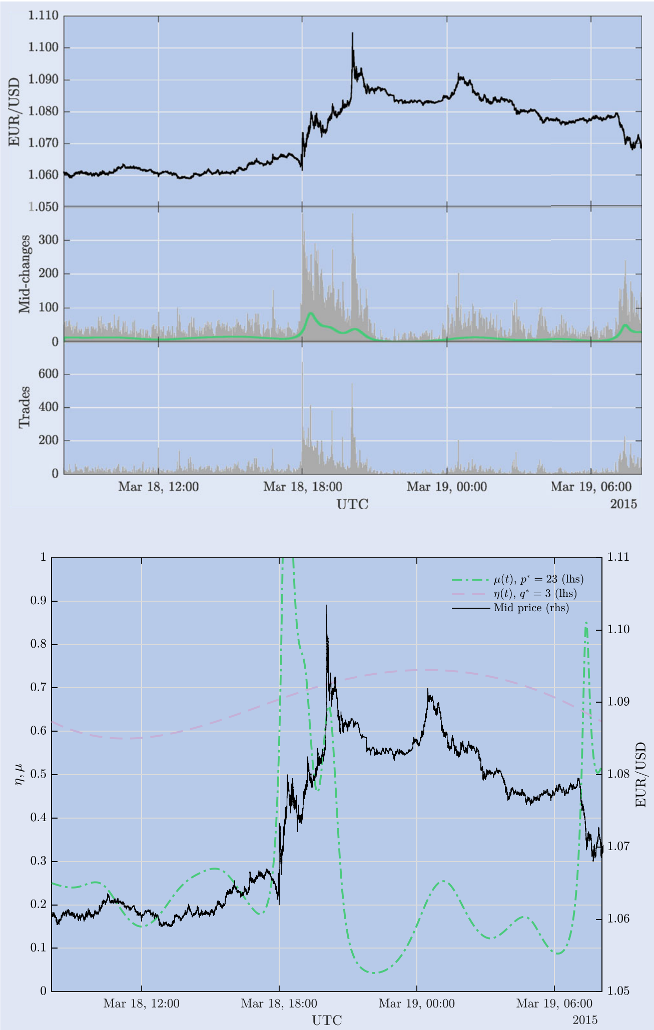


Figure 4. Same as figure 2 for the EUR/USD event on 18 March 2015. The bottom panel shows estimates based on randomized timestamps ($\Delta = 100$ milliseconds).

by highly elevated trading and quoting activity on the foreign exchange spot market. In addition, Euro foreign exchange futures contracts on the CME triggered a circuit breaker which resulted in the price being stuck at some level for several minutes,† leading the Euro foreign exchange futures price to being artificially held back relative to other currency futures.‡ Since foreign exchange futures and spot markets are tightly coupled through arbitrage relations, we can expect such dislocations in the futures market to also propagate to spot markets if they are persistent enough. Looking at the estimates of the selected Hawkes(23,3) model in figure 4, we see first that the estimator for the exogenous intensity spikes in excellent agreement with the release of the FOMC statement and the incident of the depreciation of the US-Dollar. The branching ratio however rises only weakly during the period, indicating a slightly higher endogenous excitation during the crash and subsequent recovery. The estimator recovers a peak value of $\eta(t) \approx 0.75$ at that time, rising from around 0.6, but remaining far from criticality. We find no difference in the dynamics when fitting on raw timestamps.

5.5. *The New Zealand dollar crash in August 2015*

On 25 August 2015, the NZD/USD cross rate dropped almost 8% over the course of just a couple of minutes. Within half an hour, the move was largely reverted. During the episode, both quoting and trading activity increased significantly (see figure 5). Similar price dynamics could also be observed in the NZD/JPY pair. Several reports suggested that a sell-off in the US equity market had triggered a wave of risk reduction and closing of carry trades, the following increase in correlation led to dealers failing to hedge cross-asset exposures and consequently withdrawing from two-sided market making, which resulted in the excessive price reaction in the NZD (Callaghan 2017, Bank for International Settlements 2017a, 2017b). The description of this event—having its source even in another asset altogether—already suggests that there were a lot of mechanisms at play that are exogenous to the price dynamics of the NZD/USD currency pair at that time. As such, it comes as little surprise that we find no

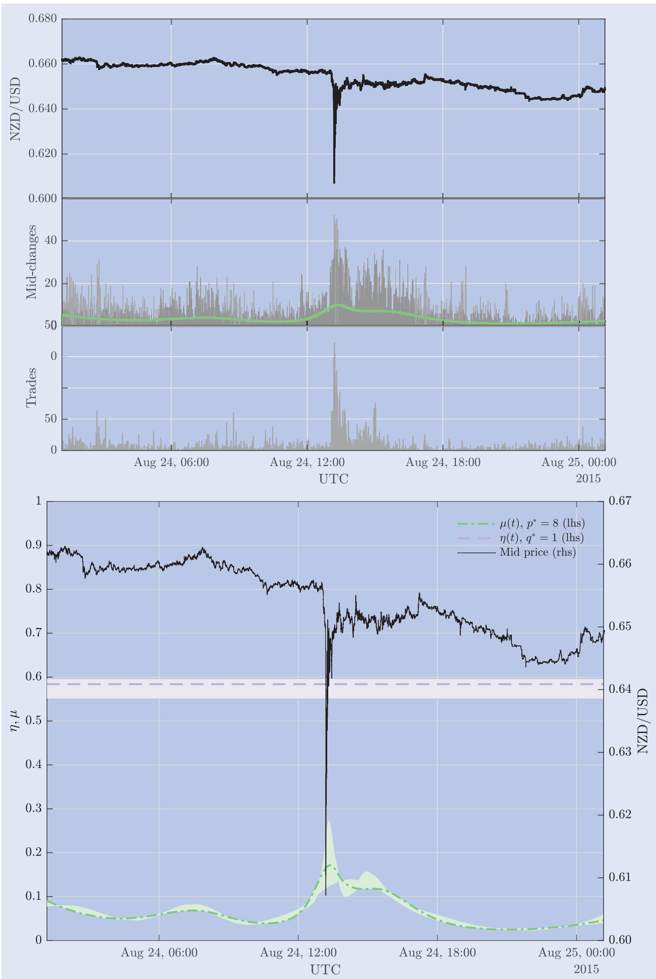


Figure 5. Same as figure 2 for the NZD/USD event on 24 August 2015. The bottom panel shows estimates based on randomized timestamps ($\Delta = 1$ millisecond).

† The CME 'Special Price Fluctuation Limits', or Rule 589 that was triggered, states that, if certain lead-month primary futures contracts trade more than some pre-determined amount outside the previous day's settlement price, a 5-minute monitoring period will be triggered. During this period, trading is only allowed at or above (below) the price limit if the price is falling (rising).
‡ See e.g. http://www.nanex.net/aqck2/4689.html for a detailed overview of the event on the futures market.

changes in the branching ratio over the event (see figure 5). The optimal Hawkes(8,1) model on randomized timestamps has constant $\eta(t) \approx 0.6$ and attests the spike in the mid-price intensity around the event to a rise in exogenous activity. Also when fitting the dynamics on the raw timestamps, the resulting optimal model is a Hawkes(8,1) process with constant endogenous dynamics of identical magnitude. We stress that our calibration finding $q^* = 1$ is encouraging evidence for the fact that—despite the significant flexibility allowed by the Hawkes(p, q) model—our approach does not bias towards always identifying some changes in the instantaneous branching ratio. The observation that a constant branching ratio is supported by the narratives of the event and is also recovered by the fits, validates our approach nicely.

5.6. The South African Rand crash in January 2016

The South African Rand—the currency with the 18th highest turnover in OTC foreign exchange according to Bank for International, Settlements (2019) and thus one of the most liquid emerging markets currencies—depreciated around 8% within just 10 minutes on 10 January 2016. Bank for International Settlements (2017a) finds that the move was triggered by retail investors unwinding carry trades, which was subsequently exacerbated by hedging of barrier options and the execution of stop-loss orders in illiquid Asian trading hours. The price dynamics during this event are reminiscent of typical dynamics in complex systems, where non-equilibrium abrupt transitions, due to exogenous shocks, are generally followed by power law relaxations (Sornette and Helmstetter 2003, Sornette *et al.* 2004a, Crane and Sornette 2008, Crane *et al.* 2010)—in particular also observed in financial time series (Lillo and Mantegna 2003, Sornette *et al.* 2004b, Weber *et al.* 2007, Petersen *et al.* 2010). The resulting Hawkes(10,4) fits† corroborate this story insofar as that the initial price drop is accompanied by a spike in the exogenous intensity, while the branching ratio only later starts to rise (see figure 6). Interestingly enough, during the subsequent relaxation/stabilization of the price move, the estimated branching ratio peaks just below criticality ($\eta(t) = 1$ is in fact within the credible interval of $[0.93, 1.01]$ at the peak) at a time when price dynamics finally seem to have calmed down—similar to what we recovered around the event in the U.S. Treasury futures market. The fact that, for these events, a rise in the endogenous dynamics towards criticality seems to have acted as a stabilizing mechanism, is reminiscent of a concept from the study of adaptive control systems, where locally adaptive stabilization is found to have the potential to move a system to critical points. In Patzelt and Pawelzik (2011, 2013) for example, it is shown that rapidly adaptive stabilization creates attractive critical points under quite general assumptions about the system dynamics. As such, we could interpret the observations around the event in the South African Rand and the U.S. Treasury futures as market participants collectively adapting to and removing predictabilities (e.g. local trends) from the price process. In the process of doing so, the

† On raw timestamps, the resulting Hawkes(9,4) fit shows almost identical dynamics, however the critical branching ratio is not within the resulting credible interval of $[0.92, 0.94]$.

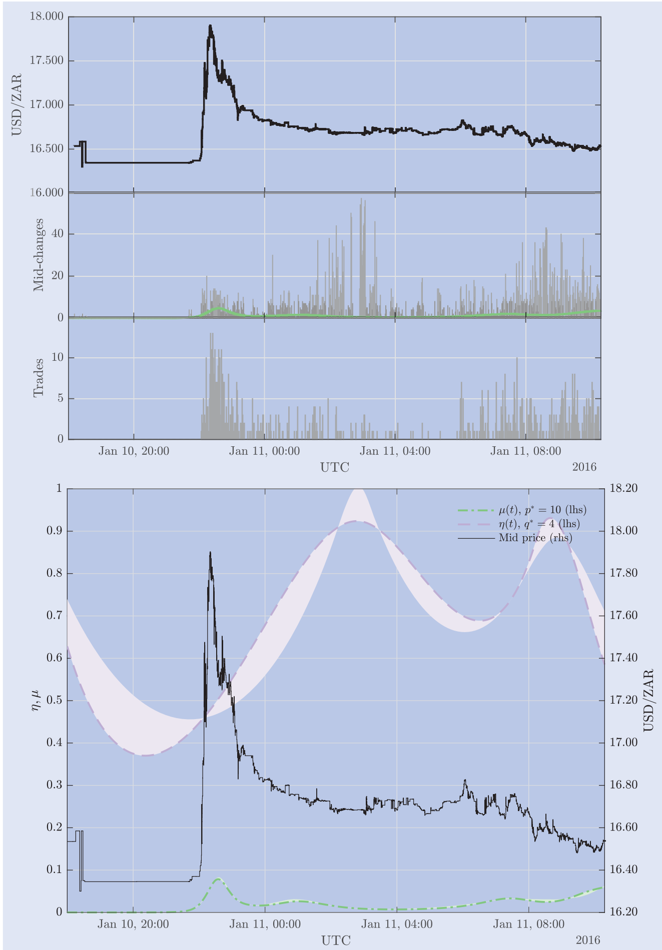


Figure 6. Same as figure 2 for the USD/ZAR event on 11 January 2016. The bottom panel shows estimates based on randomized timestamps ($\Delta = 1$ millisecond).

market indeed self-organized temporarily towards a locally critical point, which we supposedly recover with our dynamic branching ratio estimator.

5.7. The British Pound crash in October 2016

Within less than a minute, on 7 October 2016, the British Pound depreciated against the US-Dollar by almost 10% and reverted most of the drop within the subsequent 10 minutes. The event remains one of the best covered ones in foreign exchange markets, with official institutions providing a meticulous account of the incident (Noss *et al.* 2017, Bank for International Settlements 2017a). The reports find that the excessive movements in one of the most liquid currency pairs in the world have resulted from a confluence of factors. First they mention the execution of a large sell order during typically illiquid trading hours, which the market apparently still absorbed in a relatively orderly manner. Several amplifying factors are then found to have contributed to a severe deterioration in liquidity and subsequent price drop. In particular, significant demand to sell Pound Sterling to hedge options positions, as the currency depreciated, appears to have played an important amplifying role. The execution of stop-loss orders and the closing-out of positions as the currency traded through key levels may also have had an impact. The

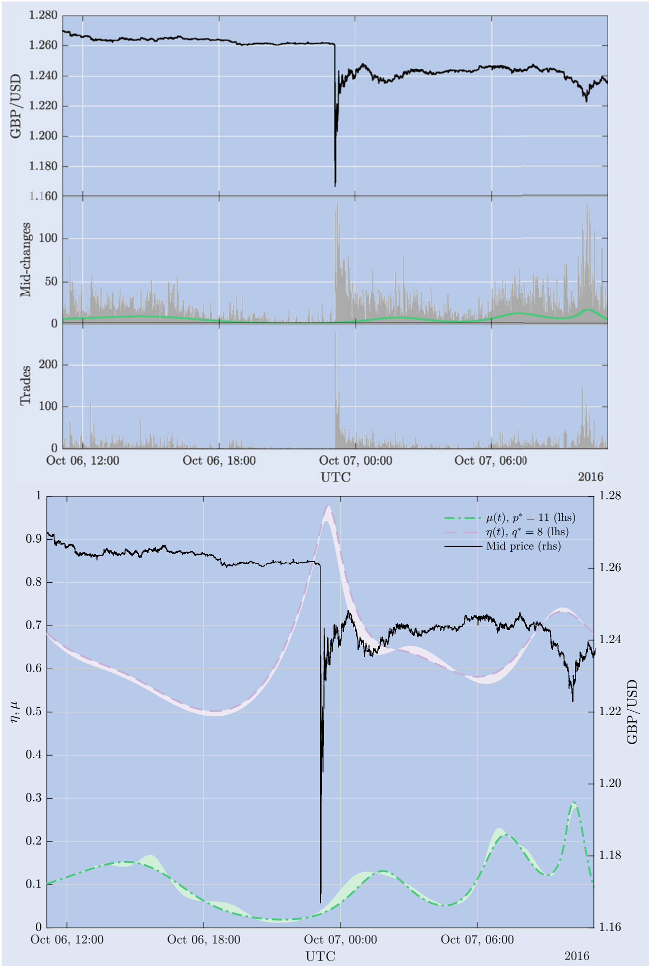


Figure 7. Same as figure 2 for the GBP/USD event on 7 October 2016. The bottom panel shows estimates based on randomized timestamps ($\Delta = 1$ millisecond).

reports also note the potentially amplifying role of circuit breakers being triggered in GBP/USD futures contracts on the CME,† which may have created larger price pressure on other trading platforms and increased the price impact of trades in the spot market. Studies of transaction level data around the event further find that dealers (primarily investment banks) were significant contributors to the liquidity dry-up during the crash. They reduced their market-making activity significantly and widened spreads, while other financial firms (such as high frequency traders and hedge funds) stepped in to provide some liquidity in their place (Schroeder *et al.* 2020).

The description of the event suggests that many of the previously discussed mechanisms that are suspected to amplify endogenous price dynamics were present during the crash—from stop-loss execution to trading halts in related markets and increased HFT participation, almost all of the ingredients for high endogeneity were apparently present. And indeed, if we look at the estimates of the optimal Hawkes(11,8) model around the event in figure 7, we find that the branching ratio rises significantly from around 0.5 to peak values of ≈ 0.98 during the price drop, reaching near-critical levels and then relaxing to average levels of 0.6 once the price dynamics stabilize. Interestingly enough, the branching ratio seems to start to increase already well before the actual drop. We suspect this leading effect to at least partly also be influenced by the fact that, at the overnight time of the day during which the event took place, trading is predominantly implemented by automated systems, typically exhibiting reflexive activity.‡ The large sell transaction may then have acted as a trigger, which through mechanistic amplifiers like stop-loss orders or trading halts evolved to the observed price drop—all of which can be expected to manifest in an elevated branching ratio. Fitting on raw timestamps yields similar dynamics represented by a Hawkes(11,6) model, albeit with a slightly lower point estimate for the peak fraction of endogenous activity of 0.91, but wider credible intervals, which in fact contain 0.98.

5.8. The Ether crash in June 2017

At 19:30 UTC on 21 June 2017 a multimillion dollar market sell order was placed on the GDAX ETH/USD order book. The extraordinary trading activity resulted in a price drop of around 30%, see figure 8. The series of market orders occurred as a single atomic operation on the matching engine, completing in less than 45 milliseconds.§ As a result, hundreds of stop loss orders and margin calls were automatically executed by the matching engine of GDAX, causing Ether to momentarily trade as low as 0.10 USD. As a precautionary measure, GDAX disabled new margin funding and trading on the ETH/USD book at 20:10 UTC. Trading resumed at 23:10 UTC. Even before the trading halt was implemented, the GDAX webcast experienced technical issues and latency due to an extreme surge in customer activity. To avoid data inconsistencies resulting from these problems, we cut off our sample at 19:53 UTC, just before the data feed became unavailable. On randomized timestamps, the optimal model is a Hawkes(19,5)¶ and reveals that the branching ratio is constantly above 0.8 and visits the critical regime twice before the occurrence of the crash, whereas it is only around 0.85 at the actual time of the price drop. To gauge how the overall level of self-excitation on the day of the crash compares to 'typical' dynamics of cryptocurrencies, we can look at the recent study of Mark *et al.* (2020), done for the Bitcoin market. They estimate a standard Hawkes on mid-price changes that were induced by market orders. At one hour sample sizes, they find $\eta \in [0.7, 0.9]$, slightly lower, but compatible with our results for Ether, which yield an average branching ratio of 0.9. The exogenous intensity in our estimates furthermore also only spikes after the price drop, which suggests that the extreme price move in fact most likely can simply be traced back to the market impact of the excessively large sell transaction. This is also corroborated by the fact that trade intensity spikes significantly, whereas quoting intensity as measured by

† Trading on the CME was halted multiple times during the event. First, for 10 seconds due to a so-called 'velocity-logic' event, then for several minutes due to reaching its daily lower price limit and later again due to a velocity-logic event (see Bank for International Settlements 2017a for more details).

‡ We have indeed found evidence for such a diurnal overnight rise in the branching ratio also in other currency pairs, see e.g. Wehrli *et al.* (2020).

§ https://medium.com/@AdamLWhite/eth-usd-trading-update-3-69e623335e00

¶ On raw timestamps, the optimal Hawkes(16,4) produces almost identical dynamics for the branching ratio.

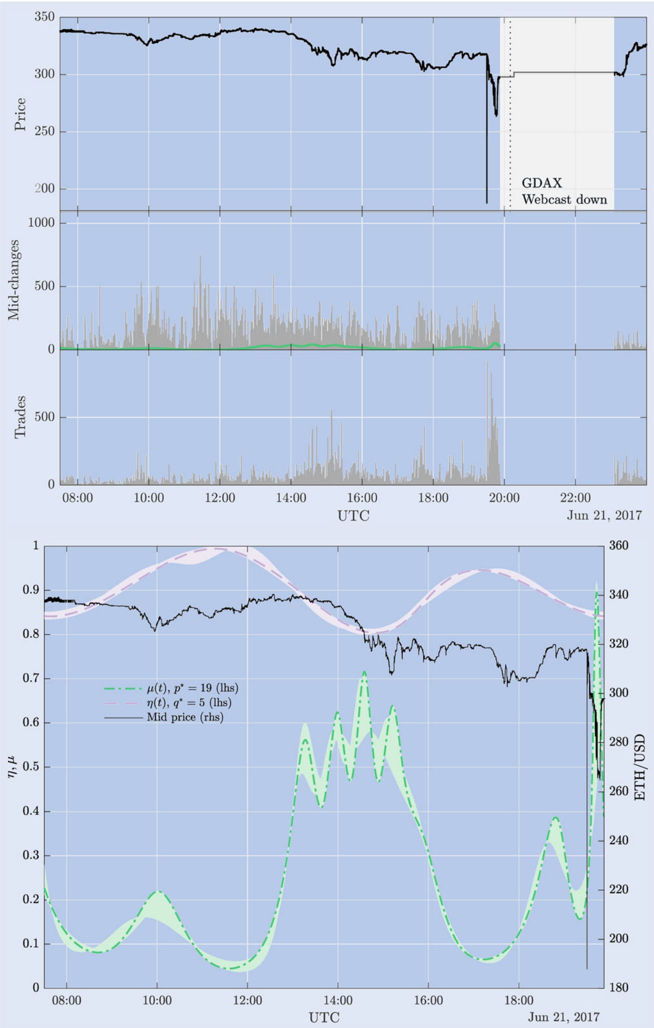


Figure 8. Same as figure 2 for the ETH/USD event on 21 June 2017. In the top panel, the time during which the GDAX data feed was down is shaded in gray, and the vertical dotted line indicates the start of the trading halt (see main text for details). The bottom panel shows estimates based on randomized timestamps ($\Delta = 1$ millisecond).

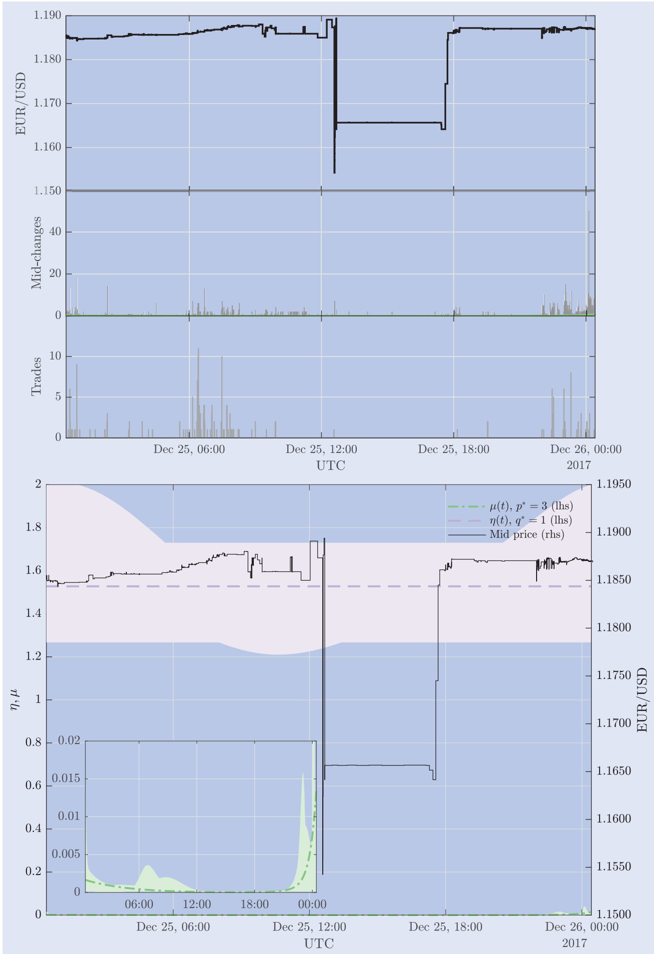


Figure 9. Same as figure 2 for the EUR/USD event on 25 December 2017. The bottom panel shows estimates based on randomized timestamps ($\Delta = 100$ milliseconds).

the mid-price jump intensity remains fairly stable around the crash.

5.9. The EUR/USD crash in December 2017

As an example of the fact that high levels of endogeneity in the price process can also be apparent around irregular price moves in situations when overall activity is extremely subdued, we look at the EUR/USD exchange rate on 25 December 2017. As can be seen from figure 9, the EUR/USD exchange rate gapped lower by around 3% instantaneously and recovered hours later to unchanged levels with very little quoting and trading activity. Even though many markets are closed during Christmas holidays, the foreign exchange market remains open for trading and prices are in fact streamed even when most staff at financial institutions are not present. Clearly, we can expect that this will lead to a lot of the live pricing decisions being made in an automated fashion—and with appropriately restrictive automated safeguards—similar to more illiquid times of the day during the European overnight times. Following this rationale, it comes as little surprise that the optimal Hawkes(3,1) model exhibits a constant and even *supercritical* level for the branching ratio at ≈ 1.5. On raw timestamps, the dynamics of the optimal Hawkes(5,1) model are comparable, with $\eta \approx 1.3$. We however also see for both fits that the credible intervals for the branching ratio, as well as the immigration are very wide—apart from model uncertainty most likely also reflecting the highly irregular distribution of inter-arrival times in this sample, which can also be expected to bias estimates of η upward† , especially if heavy-tailed offspring densities, like the approximate power law we rely on here, are used. Overall, the activity patterns and estimates around the event suggest that the supposed flash crash in this instance was more likely just some market makers stopping from quoting on EBS for some time or widening spreads significantly in subdued holiday trading, to the point where the mid-price dislocated downward.

5.10. The USD/JPY crash in January 2019

One of the most recent events in foreign exchange concerns the Japanese Yen on 2 January 2019. In just 30 seconds, the

† For a discussion regarding the effect of outliers on branching ratio estimates, see Filimonov and Sornette (2015b).

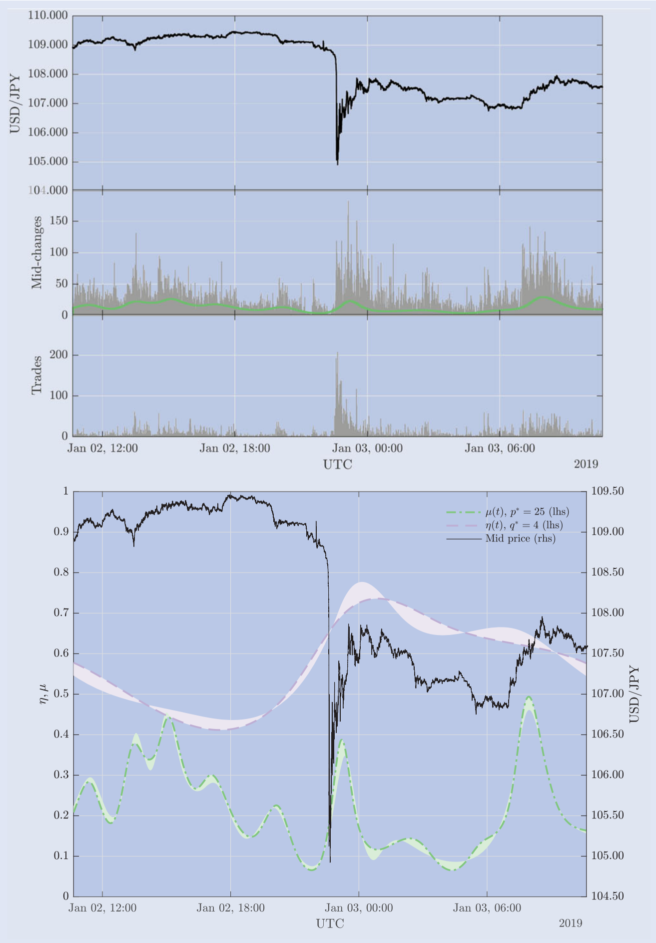


Figure 10. Same as figure 2 for the USD/JPY event on 2 January 2019. The bottom panel shows estimates based on randomized timestamps ($\Delta = 100$ milliseconds).

Yen appreciated by 3% against the US-Dollar, a comprehensive description of the event can be found in Reserve Bank of Australia (2019). Three key factors are found to have contributed to the deterioration in market conditions. First, there were significant liquidations of carry positions by Japanese retail investors (see also Han and Westelius 2019). Because of daily system halts at retail aggregators, clients were not able to post additional margin or close positions to avoid triggering automatic stop-losses. Failure to post additional margin resulted in an automatic liquidation by brokers of the positions. Second, liquidity was seasonably reduced at the time, due to the time of year and day, but also since Japanese banks were additionally closed for a public holiday. Third, the safety mechanisms of automated market making systems may have acted as amplifiers during the event as they automatically switched off when market conditions turned unusual. Further exacerbating all these facts might be that CME futures markets were closed at the time, making it more difficult to hedge positions, particularly if they were one-sided.

Looking at our model estimates for the event in figure 10, we can see that the branching ratio indeed rises leading up to and during the crash, in line with the endogeneity-inducing factors like stop-loss executions reported to play a role in the crash. However, even at the peak, the branching ratio from the optimal† Hawkes(25,4) model stays well away from criticality, with a maximum of ≈ 0.75. The exogenous intensity also spikes during the event, but only after the trough of the crash, during the initial rebound of the price.

6. Residual analysis and refined treatment of exogenous clustering

In order to address the basic adequacy of our Hawkes(p, q) model to describe the univariate mid-price microstructure around flash crashes, we perform residual analysis and a comparison with alternative models.

6.1. *Residual analysis*

The standard way to assess the quality of point process fits also applies to the Hawkes(p, q) model and is based on the standardized residual process $\xi_i = \int_0^{t_i} \lambda(t|\hat{\boldsymbol{\theta}})\mathrm{d}t$ with corresponding increments $r_i = \xi_i - \xi_{i-1}$. Under the null hypothesis that the t_i were generated by the Hawkes(p, q) process, the residual process ξ_i should be a Poisson process with unit intensity (Cinlar 2013). Given the fact that we attempt to fit days with particularly extreme realizations of non-stationary price behaviors, with tens of thousands of points, and with a randomization procedure to correct for limited timestamp resolution and coinciding points, it is not reasonable to assume that our still fairly simple model (1) will be able to fully explain a complete day of complex and diverse trading activity in a period around a flash crash. As such, it is unsurprising that, e.g. a Kolmogorov–Smirnov test confidently rejects the null hypothesis of unit exponentiality of r, except for the event in USD/ZAR. The deviations from the null hypothesis of the unit exponential distribution hereby occur predominantly for very short durations. For our purpose of explaining dynamics in exogenous and endogenous activity, the precise distribution of the residuals is however not essential. Rather, we are interested in (i) trends and (ii) autocorrelation being described sufficiently well by our model. For (i) we can check the uniformity of the ξ_i on $(0, N_T]$, e.g. by plotting a histogram. (ii) can be evaluated with standard tests for significant autocorrelations, here we choose to complement a visual check of the ACF with a Ljung-Box test (Ljung and Box 1978).

Looking at the summary of the residual checks in figures 11 ($\Delta = 0$) and 12 (randomized timestamps), we can see that the Hawkes(p, q) model does a good job in achieving approximate uniformity of the residuals. Thus, trends are successfully removed by our flexible treatment of $\mu(t)$. Regarding remaining significant autocorrelations, we find that on raw/unique timestamps ($\Delta = 0$) the model indeed captures a wide variance of the event rate and in general very little residual

† The optimal model on raw timestamps has (20,8) degrees of freedom and a peak branching ratio of 0.78 at 00:00 UTC. Since $p^* = 25$ on pre-processed timestamps is at the border of the explored range, we have additionally extended the model search up to $p = 35$. Using the extended search, the optimal p^* is 27 and q^* remains unchanged at 4. Also the estimates of the branching ratio around the flash crash remain identical, only the model uncertainty is reduced.

Figure 11. Goodess of fits on raw/unique timestamps. Shows the sample autocorrelation function $\hat{\rho}_k$ of the residuals $r_i = \xi_i - \xi_{i-1}$ from Hawkes(p, q) fits on timestamps without randomization $(\Delta = 0)$. The horizontal dashed lines are 95% approximate confidence intervals for a white noise process. The ξ_i are obtained from integrating the conditional intensity of the optimal fit up to t_i. Each plot includes the p-values from a modified Ljung–Box test, where the first-order autocorrelation is excluded from the test, i.e. the test statistic is $S = N_T(N_T + 2)\sum_{k=2}^{h+1} \frac{\hat{\rho}_k^2}{N_T - k} \sim \chi^2(h)$. In the insets, we additionally plot histograms of the standardized residuals ξ on $(0, N_T]$.

correlation remains. But, there are some significant autocorrelations present at short lags, leading to confident rejection of uncorrelated residuals r and squared residuals r^2 for the majority of events—even when the broadly present, first lag negative autocorrelation is excluded from the Ljung–Box test. This first-order feature of the residual ACF is indeed particularly noticeable and warrants a more detailed investigation below. On randomized timestamps, we notice a significant amount of autocorrelation remaining for the E-mini and U.S. Treasury data sets, where duplicates were present before randomization. While the magnitudes of the remaining autocorrelations are rather small, the goodness of fit for the two futures data sets, even for the best fitting assumption of $\Delta = 1$ second presented here, is clearly inferior to the ones on the foreign exchange and cryptocurrency data—where after randomization the goodness of fit is comparable to when using raw timestamps. These results clearly highlight the importance of high-quality timestamps in the application of financial point processes. In fact, we find that, when duplicates are present, the assumptions on Δ have a significant effect on the goodness of fit as measured by the residual autocorrelation. In line with this result, the choice of Δ also has a clear effect on the estimates of the offspring density parameters (see table 3). In particular, we observe that both the mode and tail index of the approximate power law tend to increase with larger randomization windows, the triggering strength according to $\langle \eta \rangle$ is however largely comparable. We note that some of these issues with data quality could be remedied by working with aggregated point processes in the form of integer-valued time series models, where the spurious microstructure induced by timestamp inaccuracies are naturally aggregated out. The integer-valued autoregressive (INAR) process representation of the Hawkes process for example and its estimation using EM algorithms, as developed in Wehrli *et al.* (2020), can be extended to time-varying branching ratios like we use here in a straightforward manner. This however goes beyond the scope of this paper and is left for future work.

6.2. *Comparison to alternative models*

Here, we return to the broadly present, negative first-order autocorrelation of the r_i uncovered in the residual analysis

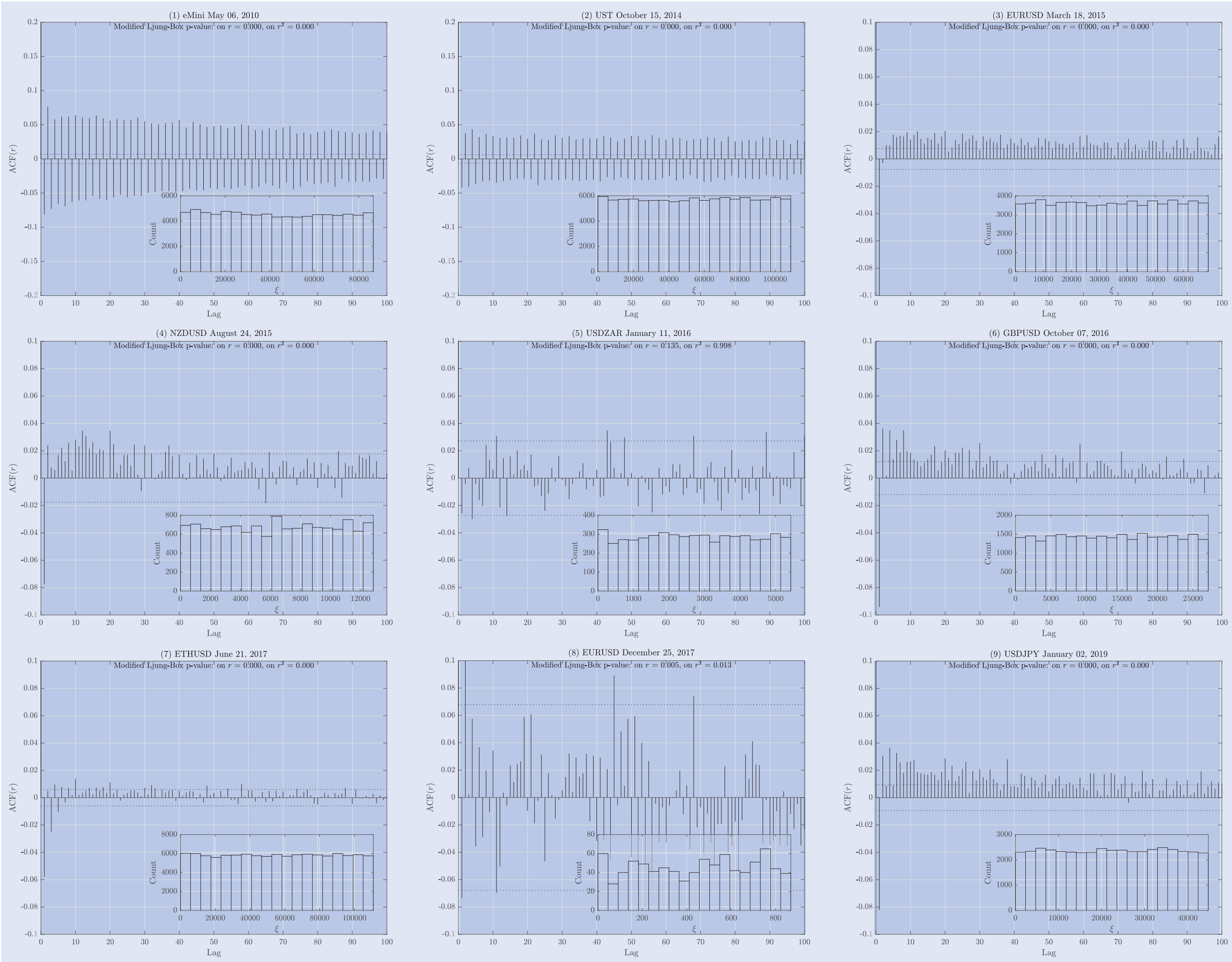


Figure 12. Goodness of fit on randomized timestamps. Shows the sample autocorrelation function $\hat{\rho}_k$ of the residuals $r_i = \xi_i - \xi_{i-1}$ from Hawkes(p, q) fits using the best fitting assumption on the timestamp uncertainty Δ. The horizontal dashed lines are 95% approximate confidence intervals for a white noise process. The ξ_i are obtained from integrating the conditional intensity of the optimal fit up to t_i. For the E-mini data and the U.S. Treasury data, the fits were performed using $\Delta = 1$ second, the remaining fits are on randomized exchange timestamps as described in the main text. Each plot includes the p-values from a modified Ljung-Box test, where the first-order autocorrelation is excluded from the test, i.e. the test statistic is $S = N_T(N_T + 2)\sum_{k=2}^{h+1} \frac{\hat{\rho}_k^2}{N_T - k} \sim \chi^2(h)$. In the insets, we additionally plot histograms of the standardized residuals ξ on $(0, N_T]$.

Table 3. Summary of the approximate power law offspring density parameter estimates (the mode τ_0 in seconds and the tail index ε) from the optimal Hawkes(p, q) fit for each event and different randomization windows Δ.

		$\Delta = 0$		$\Delta = 1$ms		$\Delta = 100$ms		$\Delta = 1$sec		$U(0, \Delta_i)$	
	Event	τ_0	ε	τ_0	ε	τ_0	ε	τ_0	ε	τ_0	ε
(1)	E-mini	0.014	0.49	10^{-4}	0.28	0.022	0.98	0.185	1.29		
(2)	U.S. Treas.	0.002	0.38					0.127	1.45	0.020	0.59
(3)	EUR/USD	0.280	0.65			0.282	0.65				
(4)	NZD/USD	3.079	0.71	3.077	0.71						
(5)	USD/ZAR	3.047	0.71	2.996	0.70						
(6)	GBP/USD	0.652	0.51	0.650	0.52						
(7)	ETH/USD	0.145	0.53	0.145	0.53						
(8)	EUR/USD	0.142	0.06			0.122	0.03				
(9)	USD/JPY	0.227	0.55			0.219	0.54				

above and take it as a motivation to compare the Hawkes(p, q) to alternative models. A first conjecture could be that this observed residual behavior is spuriously introduced by the randomization procedure. However, fits on raw timestamps also exhibit comparable residual ACF and we thus must assume that it is a genuine feature of the data set, which the process (1) is not able to capture. As an alternative hypothesis, we could think that the residual first-lag dependence is

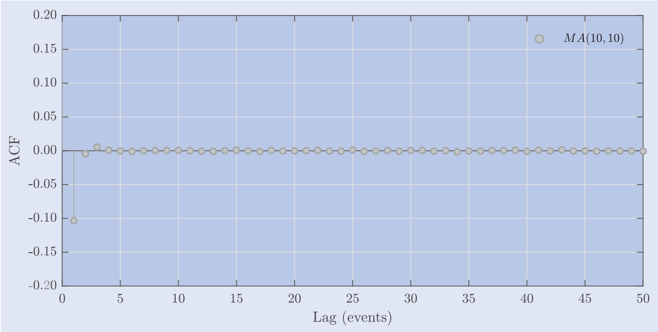


Figure 13. Shows the average empirical autocorrelation function of the inter-arrival times from 100 simulations of a moving-average point process (a shot-noise or Neyman–Scott process) with conditional intensity (4). The kernel is of the form $\Theta(t) = \alpha e^{-\beta t}$ and the process thus denoted MA(α,β). The standard errors of the average autocorrelations across the simulations are extremely small and not shown here.

an indication that the Hawkes(p, q) process—or more particularly our estimator for $\mu(t)$—does not capture a moving-average type feature of the data, which in the context of point processes corresponds to a shot-noise type of exogenous clustering. To support this intuition, we simulate from a shot-noise Cox process (Møller 2003) with conditional intensity

$$\lambda^*(t) = \mu + \int_{-\infty}^{t} \Theta(t-s)\mathrm{d}N_s^{\mu}, \tag{4}$$

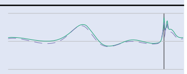
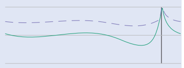
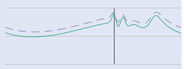
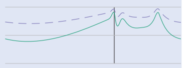
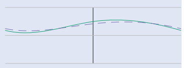
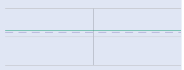
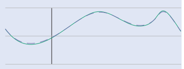
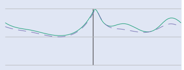
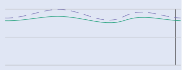
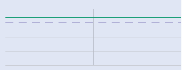
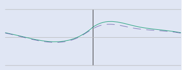
which is the intensity of a process N^*, driven by a homogenous Poisson process N^{μ} with intensity $\mu > 0$. The exogenous clustering in this process is described by the kernel $\Theta : [0,\infty) \to [0,\infty)$ with $\gamma = \int_0^{\infty} \Theta(t)\mathrm{d}t \in [0,\infty)$, so that $f(t) = \Theta(t)/\gamma$ is a density with primitive $F(t) = \int_0^t f(s)\mathrm{d}s$. The resulting ACF of the inter-arrival times is given in figure 13 for the particular case of an exponential density.

The negative first-order dependence we are looking for is apparent, suggesting that an extension of the Hawkes(p, q) model in the spirit of the autoregressive moving-average (ARMA) point process introduced in Wheatley *et al.* (2018) to

$$\lambda(t) = \mu(t) + \int_{-\infty}^{t} \Theta(t-s)\mathrm{d}N_s^{\mu} + \int_{-\infty}^{t} \eta(s)h(t-s)\mathrm{d}N_s \tag{5}$$

might provide a more comprehensive treatment of exogenous clustering than simply relying on flexible $\mu(t)$ to capture such bursts. In addition, this allows us to test whether the

Table 4. MA-Hawkes(p, q) fits with $\Theta(t) = \gamma\beta e^{-\beta t}$ and approximate power law offspring for the Hawkes kernel.

#	Event	Δ	p^*	q^*	$\langle\mu\rangle$	γ	β^{-1}	$\langle\eta\rangle$	p-value	$\eta(t)$
(1)	E-mini (06/05/2010)	0	17	18	0.299 (0.342)	0.01	0.018	0.56 (0.54)	0.854	
		1 sec	19	7	0.246 (0.299)	1.26	0.409	0.50 (0.74)	0.990	
(2)	U.S. Treasury (15/10/2014)	0	18	12	0.149 (0.153)	0.39	0.247	0.63 (0.71)	0.979	
		1 sec	21	12	0.148 (0.178)	1.55	0.493	0.58 (0.80)	0.990	
(3)	EUR/USD (18/03/2015)	100 ms	23	3	0.220 (0.262)	0.16	0.297	0.67 (0.67)	0.958	
(4)	NZD/USD (24/08/2015)	1 ms	8	1	0.058 (0.062)	0.00	17.259	0.61 (0.58)	1.000	
(5)	USD/ZAR (11/01/2016)	1 ms	10	4	0.018 (0.019)	0.01	32.822	0.68 (0.68)	0.978	
(6)	GBP/USD (7/10/2016)	1 ms	11	8	0.090 (0.103)	0.01	1.204	0.68 (0.64)	0.990	
(7)	ETH/USD (21/06/2017)	1 ms	19	5	0.179 (0.247)	1.50	0.085	0.82 (0.90)	0.991	
(8)	EUR/USD (25/12/2017)	100 ms	3	1	<0.001 (0.001)	0.41	0.173	1.69 (1.53)	0.622	
(9)	USD/JPY (02/01/2019)	100 ms	25	4	0.198 (0.218)	0.04	0.251	0.60 (0.57)	0.881	

Notes: The MCEM estimation is done with $K = 50$ samples taken uniformly from the second half of a chain of length 100 000, generated by the Metropolis–Hastings algorithm. The p-values are from a likelihood ratio test for the null of the Hawkes(p, q) model against the alternative of the corresponding MA-Hawkes(p, q), performed on an MCMC sample of $K = 100$ draws from a Markov chain of length 300 000. The values in parentheses denote the respective estimates from the Hawkes(p, q) model and the units of the average immigration $\langle\mu\rangle$ are sec^{-1} and the MA kernel time scale β^{-1} is expressed in seconds. The column $\eta(t)$ provides sparklines for the branching ratio function estimates of the respective MA-Hawkes(p, q) fit in solid green and the corresponding Hawkes(p, q) fit in dashed purple, where the functions are plotted on $[0, T]$ and the three horizontal grid lines indicate levels for $\eta(t)$ of 0, 0.5 and 1, respectively. The vertical black line indicates the trough/peak of the flash crash.

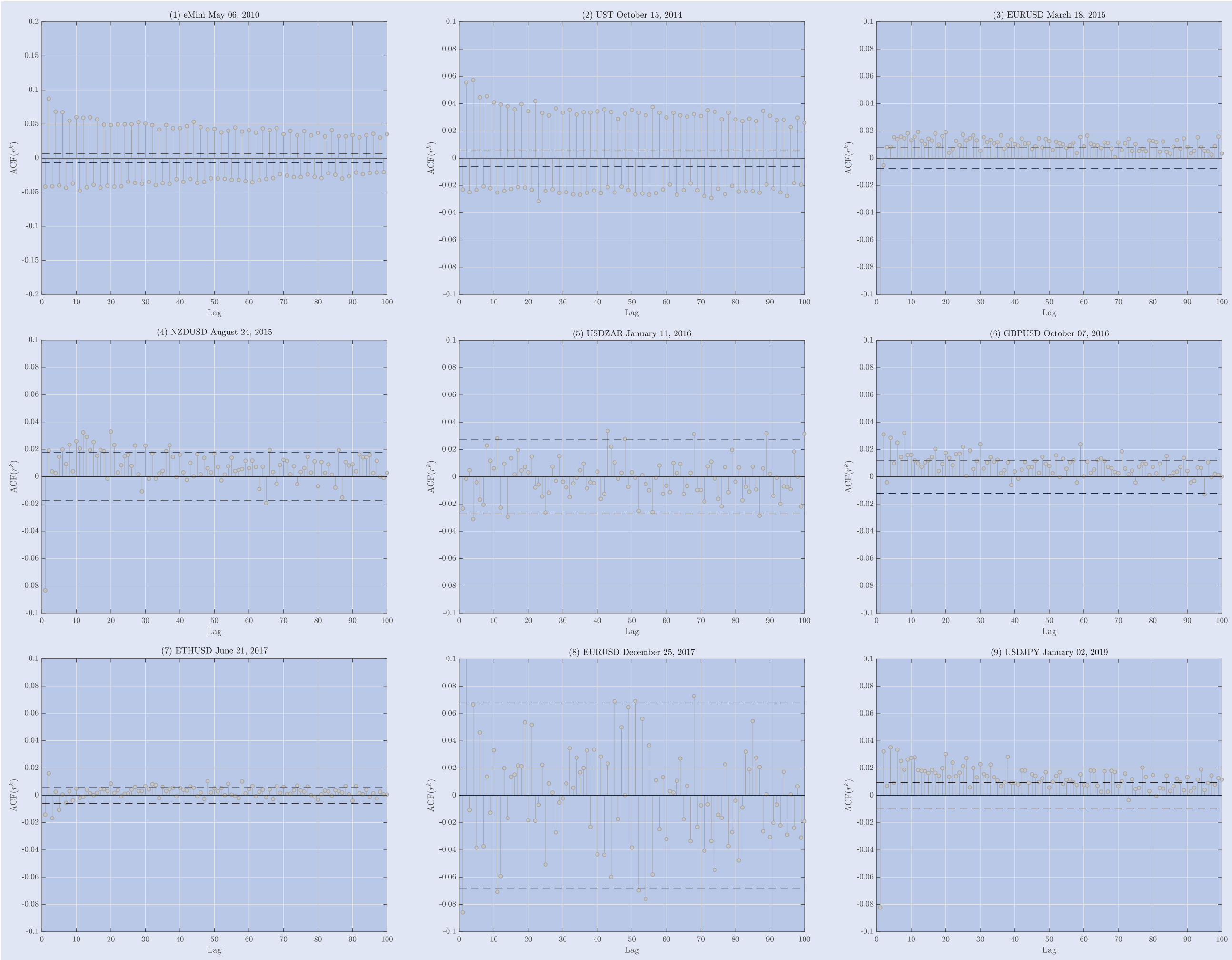


Figure 14. Shows the average sample autocorrelation function of the residuals r^k obtained by MCMC simulation of the immigrant process, conditional on the parameter estimates of the process (5). The average is constructed by drawing a realization of the immigrant process $K = 100$ times from a Markov chain of length 300 000, generated by the Metropolis–Hastings algorithm, and then computing the empirical ACF of the resulting residuals $r_i^k = \xi_i^k - \xi_{i-1}^k$ according to (7). The horizontal dashed lines indicate 95% approximate confidence intervals for a white noise process. The standard errors of the mean autocorrelations at each lag across the MCMC draws are extremely small (< 0.001) and not shown here. The MA-Hawkes(p, q) processes were produced using the best fitting assumption on the timestamp uncertainty Δ, see main text for details.

point clusters in the data are better explained by Hawkes type (endogenous) or shot-noise Cox type† (exogenous) clustering, which has particular relevance when assessing a hypothesis of criticality.

Because the process with conditional intensity (5) is an extension of the ARMA point process, it can be estimated using a Monte Carlo Expectation Maximization (MCEM) algorithm (Wei and Tanner 1990), where Markov Chain Monte Carlo (MCMC) sampling through a Metropolis–Hastings algorithm is used to perform conditional simulation of the immigrant process in the E-step (Wheatley *et al.* 2018). A description of the algorithm to estimate (5) is given in Appendix 3.

A significant practical difficulty of the process (5) is that, because the immigrants N^{μ} are unobserved, both residuals and unconditional likelihood are inaccessible for testing and model selection. We solve the latter here by assuming that the optimal number of degrees of freedom from the Hawkes(p, q) fits are also optimal for the extended 'MA-Hawkes(p, q)' model (5) and fix them for estimation. Since we can expect that the inclusion of additional structure for exogenous clustering through $\Theta(t)$ will—if anything—reduce the required flexibility in $\mu(t)$, it seems reasonable to assume that the optimal extended model would require $p \leq p^*, q \leq q^*$. As making the models for μ too flexible usually does not introduce significant bias in the estimated average levels of endogenous excitation (see also Wheatley *et al.* 2019 and Wheatley *et al.* 2018), we propose that this approach can be used to circumvent the inaccessibility of the unconditional likelihood for the present purpose. To assess the significance of the improvement from including $\Theta(t)$, we estimate likelihood ratios using MCMC as discussed in Møller and Waagepetersen (2003). In particular, the likelihood ratio for some Hawkes(p, q) parameter estimate $\hat{\boldsymbol{\theta}}$ and a corresponding MA-Hawkes(p, q) process with estimates $\hat{\boldsymbol{\theta}}_1$ is approximated by

$$r(\hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\theta}}_1) := \frac{p(\boldsymbol{\omega}|\hat{\boldsymbol{\theta}})}{p(\boldsymbol{\omega}|\hat{\boldsymbol{\theta}}_1)} \approx \frac{1}{K} \sum_{i=1}^{K} \frac{p(\boldsymbol{\omega}, \boldsymbol{c}^{(i)}|\hat{\boldsymbol{\theta}})}{p(\boldsymbol{\omega}, \boldsymbol{c}^{(i)}|\hat{\boldsymbol{\theta}}_1)}, \tag{6}$$

† Note that the particular specification of the dynamics for the shot-noise contributions $\Theta(\cdot)$—having unit marks—as in (5) specializes the shot-noise Cox process to a modification of the well-known Neyman–Scott process (Neyman and Scott 1958).

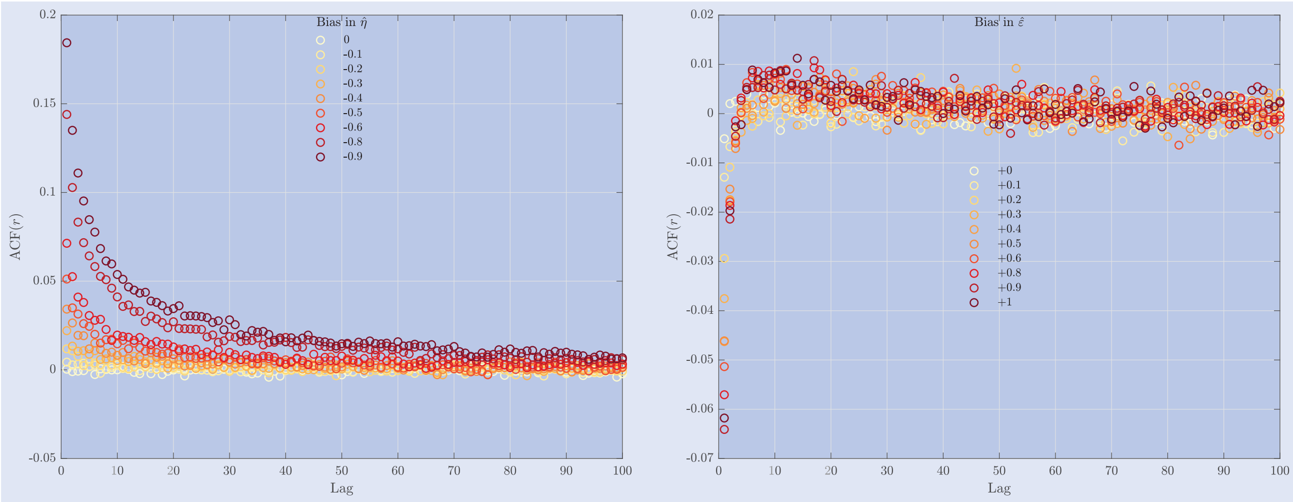


Figure 15. Shows the average autocorrelation function of the residuals $r_i = \xi_i - \xi_{i-1}$, depending on the bias of the parameter estimates for the branching ratio (left panel) and the tail index of the approximate power law kernel (right panel). The residuals are obtained by simulating a Hawkes(1,1) process and computing the corresponding residual process using parameters from a model where all but one parameter are kept at their true values, but the focal parameter is shifted according to the bias indicated in the legends of the plots. The immigration is fixed to a constant of $\mu = 0.1$ throughout and the kernel is parametrized as an approximate power law with $\tau_0 = 0.1$. In the left panel, the tail index is fixed at a value of $\varepsilon = \hat{\varepsilon} = 0.5$, the true branching ratio is set to $\eta = 0.9$ and the branching ratio of the artificial estimator is varied from $\hat{\eta} = 0.9$ to $\hat{\eta} = 0$. In the right panel, the same procedure is performed for the tail index of the kernel. Now the branching ratio is fixed at $\eta = \hat{\eta} = 0.8$, the true tail index is still $\varepsilon = 0.5$ and the artificial estimator of the tail index is varied from $\hat{\varepsilon} = 0.5$ to $\hat{\varepsilon} = 1.5$. Each combination is simulated 50 times.

where $\boldsymbol{c}^{(1)}, \boldsymbol{c}^{(2)}, \ldots, \boldsymbol{c}^{(K)}$ is an MCMC sample of immigrant realizations with density $\propto p(\boldsymbol{c}|\boldsymbol{\omega}, \hat{\boldsymbol{\theta}}_1)$—just like in the E-step of the MCEM algorithm (see Appendix 3)—being the density of an immigrant realization $\boldsymbol{c}$, conditional on the parameter estimates $\hat{\boldsymbol{\theta}}_1$ and the data $\boldsymbol{\omega}$. Once the approximate likelihood ratio is computed on a sufficiently large MCMC sample, standard hypothesis tests are available since $-2\log r(\hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\theta}}_1) \dot{\sim} \chi^2(1)$, and we can thus also directly test the null of the Hawkes(p, q) model against the alternative of the MA-Hawkes(p, q).

To further assess the goodness of fit of the extended model without having access to the true residual process, we again resort to conditional MCMC simulation. Each draw $\boldsymbol{c}^{(k)}$, $k = 1, \ldots, K$ from a simulated Markov chain of immigrants is used to compute a 'conditional' residual series

$$\xi_i^k = \int_0^{t_i} \mu(t)\mathrm{d}t + \sum_{j:c_j^k < t_i} \gamma F(t_i - c_j^k) + \sum_{j:t_j < t_i} \eta(t_j) H(t_i - t_j), \tag{7}$$

where $F(t) = \int_0^t f(s)\mathrm{d}s$ and $H(t) = \int_0^t h(s)\mathrm{d}s$ are, as before, the primitives of the exogenous and endogenous offspring densities. On these conditional residuals, the standard tests mentioned above apply. One can then perform these tests on each draw from the Markov chain and look at distributions of the test results over the ensemble of performed draws.

We present results from fitting the MA-Hawkes(p, q) in table 4, where we have used $\Theta(t) = \alpha e^{-\beta t}$, so that the exogenous branching ratio is given by $\gamma = \alpha/\beta$, and as before use an approximate power law offspring density for the Hawkes kernel. Overall, the estimates suggest only very weak exogenous clustering effects apart from what our flexible estimator for $\mu(t)$ already covers. At most, we find an exogenous clustering coefficient of around 1.5, but mostly γ is in fact close to zero and the corresponding characteristic times are rather short. With the estimated exogenous clustering coefficient γ being less than 2, the shot-noise process is providing near-Poissonian immigration to drive the Hawkes processes and thus the MA-Hawkes(p, q) is not an obvious replacement for the standard Hawkes(p, q) model. This is also formally confirmed by p-values of the performed likelihood ratio tests failing to reject the null of the Hawkes(p, q) for all events. In line with the Hawkes(p, q) prevailing over the MA-Hawkes (p, q) in a formal test, the average autocorrelation of the simulated residuals in figure 14 also shows that the short-lag negative autocorrelation remains in most cases (except where $\gamma > 1$ it is visibly reduced, e.g. for ETH/USD), and the overall goodness of fit according to this measure is not improved significantly. Regarding the dynamics of $\eta(t)$ recovered from the MA-Hawkes(p, q) fits, we observe that, while the average level of η is lower for larger γ, its peak values at crash times remain largely unchanged when additional shot-noise effects are modeled.

A third hypothesis for the negative short-lag residual pattern in our Hawkes(p, q) fits is that the estimated approximate power law kernel fails to describe the true offspring kernel dynamics in some respect, which then manifests in the observed residual autocorrelation. To test this hypothesis, we evaluate the effect of a bias in the parameter estimates of the approximate power offspring density on the residuals in a simulation study, for the simplified case of a Hawkes(1,1) model. We simulate realizations for a set of kernel parameters and then compute the residual process for these synthetic realizations, based on a parameter set where one focal parameter is increasingly biased away from the true value. From figure 15, we can see that underestimating the branching ratio—as expected—leads to increasingly positively autocorrelated residuals, where the autocorrelation is more pronounced and persistent, the more downward bias is introduced. Clearly, this is not the pattern we observe in our empirical fits and we thus conclude that a bias in the estimates for $\eta(t)$ can be ruled out. However, when the tail index of the approximate power law is overestimated, we indeed observe a pattern in the residual autocorrelation that is very similar to what can be observed for the real data sets in Figures 12 and 14, with negative autocorrelation at very short lags that increases in magnitude as the bias in the tail index grows.† In fact, the presence of this upward bias for the the tail index in our estimator was already documented in the simulation study in section appendix 2. Naturally, such an effect can also be expected to arise, if the true empirical kernel shape simply does not exactly follow the power law dynamics we impose here.

As such, we conclude that, while our Hawkes(p, q) estimator with approximate power law memory does not perfectly describe the shape of the empirical offspring density $h(t)$, it seems to satisfactorily encode both time-varying exogenous activity $\mu(t)$ and endogenous triggering strength $\eta(t)$, thus capturing the main features of the data, which is indeed what our objective was for the present study.

7. Conclusion

We have presented a cross-asset analysis of self-excited dynamics around events with highly irregular and cascading price dynamics, commonly referred to as flash crashes. Using a novel extension of the class of self-excited conditional Poisson processes with time-varying endogenous feedback effects—the Hawkes(p, q) process—we document instantaneous characteristic signatures of potentially anomalous market regimes around these flash crashes. Systematically balancing the flexibility of estimators for exogenous (p) and endogenous (q) activity sources, we find that, out of the nine events studied here, five exhibit a precursory rise of the branching ratio, of which three reached (near-)critical levels of self-excitation. On the basis of our exemplary selection of events, we find that the events in equity and bond futures are both to be classified to be of endogenous nature, whereas only few of the foreign exchange crashes are found to fall in the endogenous category and many are in fact triggered exogenously. The fact that flash crashes in foreign exchange more often appear to be of exogenous nature is in line with the fact that, at time scales up to a few days, we generally find higher levels of endogeneity in equity futures than foreign exchange markets (Wehrli *et al.* 2020), which are naturally more exposed to shocks from international news and macroeconomic releases. In some cases, the abnormal price moves furthermore seem to be a simple consequence of a large transaction instantaneously exhausting available liquidity. To improve identification in such situations where trading activity might have acted as a trigger, quote and trade arrivals could also be modeled jointly. For this purpose, the Hawkes(p, q) model—just like the standard Hawkes process—can be extended to a multivariate form, where all the self- and cross-triggering strengths are time-dependent. Clearly, this will result in a highly complex model and can be expected to be computationally demanding to apply to large data sets. In the multivariate setting, taking the necessary effort, the EM algorithm may however be even more useful than in the univariate case, as it decomposes and renders tractable estimation that would otherwise become highly problematic, with far too many parameters for reliable direct numerical MLE.

Regarding broader methodological aspects of financial point process modeling, we also highlight that by systematically selecting models for $\mu(t)$ and $\eta(t)$, we consistently recover more exogenous than endogenous heterogeneity in the mid-price dynamics. This means that, even though q^* is frequently bigger than one, we overwhelmingly obtain $p^* > q^*$ in our estimates, providing additional evidence for the paramount necessity to control for exogenous heterogeneity in financial point process estimation, as argued in Wheatley *et al.* (2019) and Wehrli *et al.* (2020). Furthermore, our extensions in the spirit of the ARMA point process have indicated that, when allowing for more rich exogenous clustering mechanisms, the branching ratio in regular times might in fact even be lower than typically documented values of 0.6–0.8, whereas peak values of $\eta(t) > 0.9$ remain in the case of supposedly endogenously triggered events. For the broader question of reflexivity in financial markets, the moving average extension that we made here is only the simplest one possible. The shot-noise term can be extended to have general shot-noise Cox bursts, with a distribution of marks that allows for the co-existence of a wide range of impacts of external shocks—in particular, so that large shocks are relatively rare. This produces a similar effect to the models from insurance risk theory that have been analyzed for example in Dassios and Jang (2003), Albrecher and Asmussen (2006), Klüppelberg and Mikosch (1995), Boumezoued (2016) and Brémaud and Massoulié (2002) and might shed even more comprehensive light on the general level of market endogeneity.

Apart from these general points, we also see some specific potential avenues for further improvement in the study of extreme financial events like flash crashes. In fact, when looking at our (MA-)Hawkes(p, q) fits, we can observe a tendency of the exogenous intensity to spike even more drastically than the branching ratio during the events. As such, through the lenses of the Hawkes(p, q) model, the explosive behavior of the empirical activity during these crashes is impossible to reproduce just by self-excitation and thus—by design—the exogenous intensity has to explain a large share of the observed dynamics. These limitations of the present formulation in explaining the level of self-excitation during flash crashes can be addressed in future work by either (i) allowing for point fertilities that follow a power law distribution

† We remark that the fact that several different mechanisms (bias in offspring density estimates and additional exogenous clustering) can produce similar patterns in second-order statistics is additional evidence for the superiority of EM-based estimation over moment-based estimators for Hawkes-type point processes. While the EM, basing its estimation on the full conditional intensity, is able to discern between true exogenous clustering (here of Neyman–Scott type) and Hawkes-type effects, an estimator on the basis of second-order statistics would find them indistinguishable.

(Saichev *et al.* 2005, Saichev and Sornette 2013) or (ii) extend the model to a non-linear process akin to Filimonov and Sornette (2011). Both approaches are complementary to a time-varying $\eta(t)$ and might alleviate some of the limitations of the current formulation.

Finally, in the spirit of cross-pollination of scientific disciplines, our method of adaptively estimating a time-dependent triggering strength might also be useful to fit nonstandard earthquake series that, e.g. appear under transient stress changes caused by aseismic forces—for instance by fluid injection—where stationary epidemic-type aftershock sequence (ETAS) models fit poorly (Kumazawa and Ogata 2014).

So why do flash crashes occur? The (perhaps unsatisfactory) answer according to our efforts here is, it depends. Our survey of flash crashes revealed that there seems to be a 'zoo' of triggers for flash crashes to occur. However, one possible theme is indeed that the presence of mechanistic factors like level-dependent hedging, automated stop-losses or circuit breakers in related markets can produce the necessary amplifying effects which result in cascading price change dynamics that are accompanied by sharply rising measures of endogenous excitation. However, we have also documented that flash crashes can happen in the *absence* of a critical price process. The fact, that there is no universality in the crash dynamics and many causes seem to exist for flash crashes to occur, is similar to what we see for big market crashes on longer time scales, as found in Johansen and Sornette (2010). Also Filimonov and Sornette (2015a)—based on an assessment of intraday financial drawdowns—find that some events can be attributed to an internal mutual-excitation between market participants, while others are pure responses to external shocks. While we document here that flash crashes seem to be accompanied by precursory rises in estimates of self-exciting activity with some empirical regularity, there is no apparent universality to the dynamics of $\eta(t)$ around the events and a flexible estimator is thus required to gain insight into the price dynamics. As such, we believe that the endo-exo classification framework we propose here can be useful post-mortem in developing remedies and better future processes, such as circuit breakers, latency floor mechanisms and other market design choices. Furthermore, it can potentially be used ex-ante for short-term forecasts in the case of the endogenously triggered class of crashes, by monitoring the feedback parameter in real-time. The design of such a surveillance system would have to consider several important points that were not addressed by the present study, such as rolling window calibration, sensitivity to window size and the elaboration of a signal detection algorithm. One particular difficulty concerning the latter, which such a potential early-warning system would have to solve based on the findings presented here, is filtering local critical points without diverging price movements. The fact that the branching ratio at times goes close to criticality intermittently without the occurrence of abnormal returns means that relying solely on the branching ratio to predict flash crashes necessarily produces false positives. A potential avenue to address these false positives could be to combine monitoring the branching ratio with some measure of liquidity risk, e.g. in the spirit of Donier and Bouchaud (2015). To make progress and validate such ideas, we think that, even though our Hawkes(p, q) framework cannot solve the problem of forecasting flash crashes on its own, it can provide a valuable forensic tool that allows building a catalog of dynamics and assessing similarities between events. This in turn can lead to the identification and validation of additional precursory measures, which when combined with the branching ratio can help bring down the type I error rate of the signal.

Finally, on a general level, our model might help understand how changes at the micro-level may affect macro-level properties of financial markets, such as the occurrence of flash crashes. If it is understood which behaviors of market participants lead to changes in the endogeneity of the price process, there will also be more clarity about which constraints can affect the overall market ecology beneficially (and adversely!). We hope that our (MA-)Hawkes(p, q) framework can provide a tool to inform such issues. A natural way to gain traction on the behavioral roots of specific endogenous activity dynamics as uncovered by our model is to construct an agent-based model of limit order book markets, where different classes of traders and their behavior can be modeled and controlled explicitly. In this theoretical/simulated setup, the contributions of the actions of each agent to subsequently measured endogenous activity dynamics could be quantified objectively and the effect of policies and market design choices explored in an experimental fashion.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Appendices

Appendix 1. The EM algorithm for the estimation of the Hawkes(p,q) process

Generally, the EM algorithm can be used to iteratively build an MLE using the *complete data likelihood function* $L^c(\boldsymbol{\theta}|\boldsymbol{\omega},\mathbf{Z})$ (McLachlan and Krishnan 2008), where $\boldsymbol{\omega} := \{t_i\}_{i=1}^{N_T}$ denotes a realization of the process N, $\boldsymbol{\theta}$ is a vector of model parameters and $\mathbf{Z}$ introduces an additional set of unobserved data, so that the estimation can be decomposed into more manageable sub-problems. For point processes with linear conditional intensities, if we were to know the branching structure of the underlying process, estimation would reduce to separate probabilistic estimations of the immigration and offspring processes (Marsan and Lengliné 2008, Veen and Schoenberg 2008, Lewis and Mohler 2011, Wheatley *et al.* 2016). As such, we define the missing data to be the branching structure of the process, represented as a lower-triangular indicator-matrix $\mathbf{Z}_{n\times n} \in \{0,1\}^{n\times n}$, where $Z_{i,j} = 1$ if point t_i is an offspring of point t_j and $Z_{i,i} = 1$ if t_i is an immigrant.

Provided an initial parameter estimate $\hat{\boldsymbol{\theta}}^{(0)}$, the EM algorithm iterates the expectation step (E-step) and maximization step (M-step) in each iteration l until $\hat{\boldsymbol{\theta}}^{(l)}$ (and the log-likelihood of the process) converges (Wu 1983, Salakhutdinov *et al.* 2003). In the E-step, the algorithm computes the expected value of the complete data log-likelihood with respect to the conditional distribution of the missing data, given the observed data and the current parameter estimates

$$Q(\boldsymbol{\theta}|\boldsymbol{\omega},\hat{\boldsymbol{\theta}}^{(l)}) = \mathbb{E}_{\mathbf{Z}|\boldsymbol{\omega},\hat{\boldsymbol{\theta}}^{(l)}}[\log L^c(\boldsymbol{\theta}|\boldsymbol{\omega},\mathbf{Z})]. \tag{A1}$$

The subsequent M-step maximizes the expected complete data log-likelihood (A1) to obtain new parameter estimates

$$\hat{\boldsymbol{\theta}}^{(l+1)} = \arg\max_{\boldsymbol{\theta}} Q(\boldsymbol{\theta}|\boldsymbol{\omega},\hat{\boldsymbol{\theta}}^{(l)}). \tag{A2}$$

The expected complete data log-likelihood of the process with conditional intensity (1) has the form

$$Q(\boldsymbol{\theta}|\boldsymbol{\omega},\hat{\boldsymbol{\theta}}^{(l)}) = \sum_{i=1}^{N_T}\left(\pi_{i,i}\log\mu(t_i) + \sum_{j:t_j<t_i}\pi_{i,j}\log\left(\eta(t_j)h(t_i-t_j)\right)\right) - \int_0^T \lambda(t)\mathrm{d}t, \tag{A3}$$

where the branching probabilities are $\pi_{i,j} = \mathbb{P}[Z_{i,j} = 1|\boldsymbol{\omega},\hat{\boldsymbol{\theta}}^{(l)}]$ and can be estimated in the E-step using random thinning (Ogata 1981) as

$$\pi_{i,i} = \frac{\mu(t_i|\hat{\boldsymbol{\theta}}^{(l)})}{\lambda(t_i|\hat{\boldsymbol{\theta}}^{(l)})} \quad\text{and}\quad \pi_{i,j} = \frac{\eta(t_j|\hat{\boldsymbol{\theta}}^{(l)})h(t_i-t_j|\hat{\boldsymbol{\theta}}^{(l)})}{\lambda(t_i|\hat{\boldsymbol{\theta}}^{(l)})}, \quad t_i > t_j. \tag{A4}$$

Using this probabilistic branching structure and, assuming that μ does not share any parameters with η and h, the M-step of the algorithm can estimate the various parameters of the immigration and offspring processes separately. First, the exogenous intensity $\mu(t)$ is estimated based on the weighted sample composed of point and weight pairs $\{(t_i,\pi_{i,i})\}_{i=1}^{N_T}$. In the simplest case of a constant immigration estimator, this is simply $\hat{\mu} = T^{-1}\sum_i \pi_{i,i}$. Regarding a particular model for $\mu(t)$, we follow the approach in Wheatley *et al.* (2019) and Wehrli *et al.* (2020) and use adaptive *logspline* models (Kooperberg and Stone 1991, 1992) with p degrees of freedom as a flexible technique to capture nuisance structures such as shocks and trends. Next, an analytical maximization of (A3) gives the objective function to maximize for the offspring density h as

$$\ell_h^c(\boldsymbol{\theta}|,\boldsymbol{\pi}) = \sum_{i=1}^{N_T}\sum_{j:t_j<t_i}\pi_{i,j}\log\left(h(t_i-t_j)\right) - \sum_{j=1}^{N_T}\eta(t_j)H(T-t_j), \tag{A5}$$

where $H(t) = \int_0^t h(s)\mathrm{d}s$ is the cumulative distribution function corresponding to the density h. Finally, to estimate $\eta(t)$, we consider the solutions $\hat{\eta}_j = \arg\max_{\eta(t_j)} Q(\theta|\boldsymbol{\omega},\hat{\boldsymbol{\theta}}^{(l)})$ as (potentially noisy) estimates for the random number of points that are triggered by events at times t_j, which are given by

$$\hat{\eta}_j = \frac{\sum_{i:t_i>t_j}\pi_{i,j}}{H(T-t_j)}. \tag{A6}$$

Estimating the full function $\eta(t)$, being the expected instantaneous number of points triggered by an event at time t, can then be done from pairs $(t_j,\hat{\eta}_j)$ using traditional regression/smoothing approaches. A straightforward approach would be to form a piecewise constant estimator by aggregating the $\hat{\eta}_j$ in some given intervals. A special case of this estimator is the constant branching ratio estimate $\hat{\eta} = \sum_i\sum_{j<i}\pi_{i,j}/\sum_i H(T-t_i)$. This approach however lacks adaptivity and a more flexible estimator is desirable to properly evaluate the dynamics of η. A first natural candidate is to perform a kernel regression using some kernel $K(\cdot)$ and estimate the branching ratio as

$$\hat{\eta}(t|\tau) = \frac{\sum_{i=1}^{N_T}\hat{\eta}_i K((t-t_i)/\tau)}{\sum_{i=1}^{N_T}K((t-t_i)/\tau)} \tag{A7}$$

for some bandwidth τ. While this approach indeed allows flexible estimation of the branching ratio, model selection (i.e. determining the bandwidth τ) is a notoriously difficult issue in nonparametric statistics and we prefer a more straightforward way to control the flexibility of our estimator. Nevertheless, this kernel estimator can be a valuable tool when doing exploratory work, for example when the influence of the flexibility of the immigration or the type of offspring density used on the estimated dynamics of η are of interest.

An alternative approach we propose here, in analogy to the treatment of $\mu(t)$, is to use some form of adaptive spline, where the degrees of freedom permitted can be controlled explicitly and balanced with the flexibility of μ. To this end, at each iteration of the M-step, we draw a Monte Carlo sample from the t_i with weights $\propto \hat{\eta}_i$. On this sample, we fit a logspline density with q degrees of freedom (knots) to obtain an optimal allocation of the knot sequence $\tilde{\mathbf{t}} := \{\tilde{t}_n\}_{n=1,\ldots,q}$. We then use these logspline-optimal knot locations to fit a piecewise cubic spline to the pairs $(t_i, \hat{\eta}_i)$ with knots at the $\tilde{t}_n$ and periodic boundary conditions, so that the spline has the desired number of degrees of freedom. This procedure has the advantage that we adaptively place knots at locations where the probability of endogenous points occurring is high and simultaneously retain the possibility to control the flexibility of the resulting function by choosing the allowed number of knots q in a similar fashion to the degrees of freedom p of the immigration. Additionally, since the logspline estimate for the immigration (log-)density is also based on cubic splines, the estimators for both endogenous and exogenous dynamics have identical flexibility and can thus be compared more easily. A final practical consideration concerns the fact that we require $\eta(t) \geq 0 \;\forall\; t$, which is not a priori guaranteed even though $\eta_i \geq 0$ by construction. A cubic spline which interpolates positive function values is not necessarily positive for all arguments—in particular this can arise when the spline needs to interpolate over regions with abrupt changes. In order to preserve non-negativity of our regression splines for $\eta(t)$, we thus choose to estimate the respective spline $\hat{s}(t|\tilde{\mathbf{t}})$ on a transformed scale, performing the regression

$$\log(1 + \eta_i) = \hat{s}(t_i|\tilde{\mathbf{t}}) + \epsilon_i, \tag{A8}$$

for some residual ϵ_i. A well-known issue in regression analysis is that retransforming a problem like (A8) back to the original metric for inference purposes can yield a biased estimator of the mean response—in our case the instantaneous branching ratio $\eta(t)$. In particular, we are interested in $\eta(t_i) = \mathbb{E}[\eta_i] = \exp(\hat{s}(t_i|\tilde{\mathbf{t}}))\mathbb{E}[e^{\epsilon_i}] - 1$ without knowledge of the distribution of e^{ϵ_i}. Our approach here is to apply a simple 'smearing factor' (Duan 1983) to approximate $\mathbb{E}[e^{\epsilon_i}]$ using $\tilde{S} := N_T^{-1} \sum_{i=1}^{N_T} e^{\epsilon_i}$ and characterize any potentially remaining bias of the resulting estimator in a simulation study presented in the sequel. The final estimator for the instantaneous branching ratio is then

$$\hat{\eta}(t) = \left(\tilde{S} e^{\hat{s}(t|\tilde{\mathbf{t}})} - 1\right)^+ \tag{A9}$$

and $(x)^+ = \max(x, 0)$ denotes the positive part, applied to remove any potentially remaining negative branching ratio values arising from the smearing estimate.

The resulting overall model candidate finally has $p + q$ degrees of freedom and we call it a Hawkes(p, q) model. To select an appropriate model, we propose in the spirit of Wheatley *et al.* (2019) and Wehrli *et al.* (2020) to use the Bayesian Information Criterion (BIC) (Schwarz 1978). Furthermore, to characterize the uncertainty inherent in the selection of p and q, we can approximate the posterior probability of some model defined by the tuple (p, q), in a Bayesian fashion (Kass and Wasserman 1995), given the data $\boldsymbol{\omega}$ using

$$P((p,q)|\boldsymbol{\omega}) \approx \frac{\exp\left(-\frac{1}{2}\Delta\text{BIC}_{(p,q)}\right)}{\sum_{m\in\mathcal{M}} \exp\left(-\frac{1}{2}\Delta\text{BIC}_m\right)}, \tag{A10}$$

where $\Delta\text{BIC}_{(p,q)}$ is the difference between the BIC of some specific model defined by (p, q) and the minimum BIC over a set of models $\mathcal{M}$ we are evaluating. Having computed the posterior probabilities for a set of models then also allows us to form credible intervals for $\mu(t)$ and $\eta(t)$ by bootstrapping percentiles of the posterior distribution over the functions by sampling with replacement from the models in $\mathcal{M}$ according to their approximate posterior probabilities (A10).

Appendix 2. Estimating transient criticality—a simulation study

We evaluate the EM-based Hawkes(p, q) estimator in a simulation study. For this purpose, we draw synthetic realizations of a process with bimodal immigration intensity $\mu(t) = 0.1 + 0.05\sin(3\pi t/T)$ and a branching ratio $\eta(t) = 0.6 + \bar{S} \cdot \Phi(t; T/4, 500)$, being constant most of the time, but with a peak described by the Gaussian function $\Phi(\cdot)$ centered on location $T/4$ and with width (standard deviation) 500. We perform two types of simulations, one where the branching ratio just touches criticality instaneously and one where it visits the critical regime for around 600 seconds and thus the divergence of the process has some time to develop. For this purpose, the Gaussian peak is scaled with a factor $\bar{S}$, so that its value at the mode is $\{0.4, 0.45\}$ and the resulting peak branching ratios are thus $\{1, 1.05\}$, resembling the hypothesized temporary rise to (or above) criticality during flash crashes. Finally, we use an offspring density like in (2) with $\tau_0 = 0.1$, $\varepsilon = 0.5$ and set $M = 15$ and $b = 5$.

Simulation and fitting is then done on 50 realizations for both types of processes considered, each on a window of length $T = 10'000$ seconds. The degrees of freedom for the immigration and the branching ratio estimator are scanned in $p, q \in [1, 10]$, where $p = 1$ and $q = 1$ correspond to the constant estimators of the immigration intensity and branching ratio, respectively. For performance, we truncate the interaction length of the memory kernel to 1000 points (Wheatley *et al.* 2019). We summarize the results of the simulation study in figure A1. Overall, we find that the Hawkes(p, q) estimator recovers the dynamics of the branching ratio quite well—in particular its mode, which will be of central interest in empirical studies. The maximum of $\eta(t)$ is underestimated marginally, where the peak branching ratio recovered on average is 0.98 when its true value is 1 and 1.01 when it is 1.05. The downward bias also extends to the the average level of $\eta(t)$, an error of ≈ 0.05 is hereby observable uniformly across $[0, T]$. The variance of $\eta(t)$ is quite small across the simulations.

Regarding the immigration intensity estimate, we observe that it is overestimated locally where the transient approach to criticality occurs, indicating more difficult separability of the dynamics when both exogenous and endogenous activity are high. This effect is naturally more pronounced when $\eta(t)$ remains in the critical regime for some time. The confidence interval also widens accordingly in the region where the endogenous dynamics spike, as endogenous and exogenous time scales become more difficult to disentangle by the EM. In the transient supercritical case, the uncertainty in estimating $\mu(t)$ is very high in this region. For single realizations, we typically observe that the uncertainty in model selection as indicated by the credible intervals around the estimate of $\eta(t)$ is higher than for the corresponding estimate of $\mu(t)$. Furthermore, we observe that the bootstrapped credible intervals for the branching ratio can be as high as 0.1, which highlights the importance to also look at the uncertainty in model selection as proposed here. Regarding the offspring density parameter estimates, we find that the high frequency cutoff τ_0 is estimated very accurately and with little variance, whereas the tail index ε is overestimated on average by ≈ 0.1—in line with the slight overall underestimation of the average level of self excitation and the applied kernel truncation—and its variance is also larger than what we observe for τ_0.

Appendix 3. The MCEM algorithm for the estimation of the MA-Hawkes(p, q) process

Here, we give a short practical description of the extended MCEM algorithm required to estimate the MA-Hawkes(p, q) process. The algorithm synthesizes the principles described in section A for the estimation of the Hawkes(p, q) model, with the MCEM algorithm as derived in Wheatley *et al.* (2018) for the inclusion of the additional shot-noise/moving-average effects in the spirit of the ARMA point process.

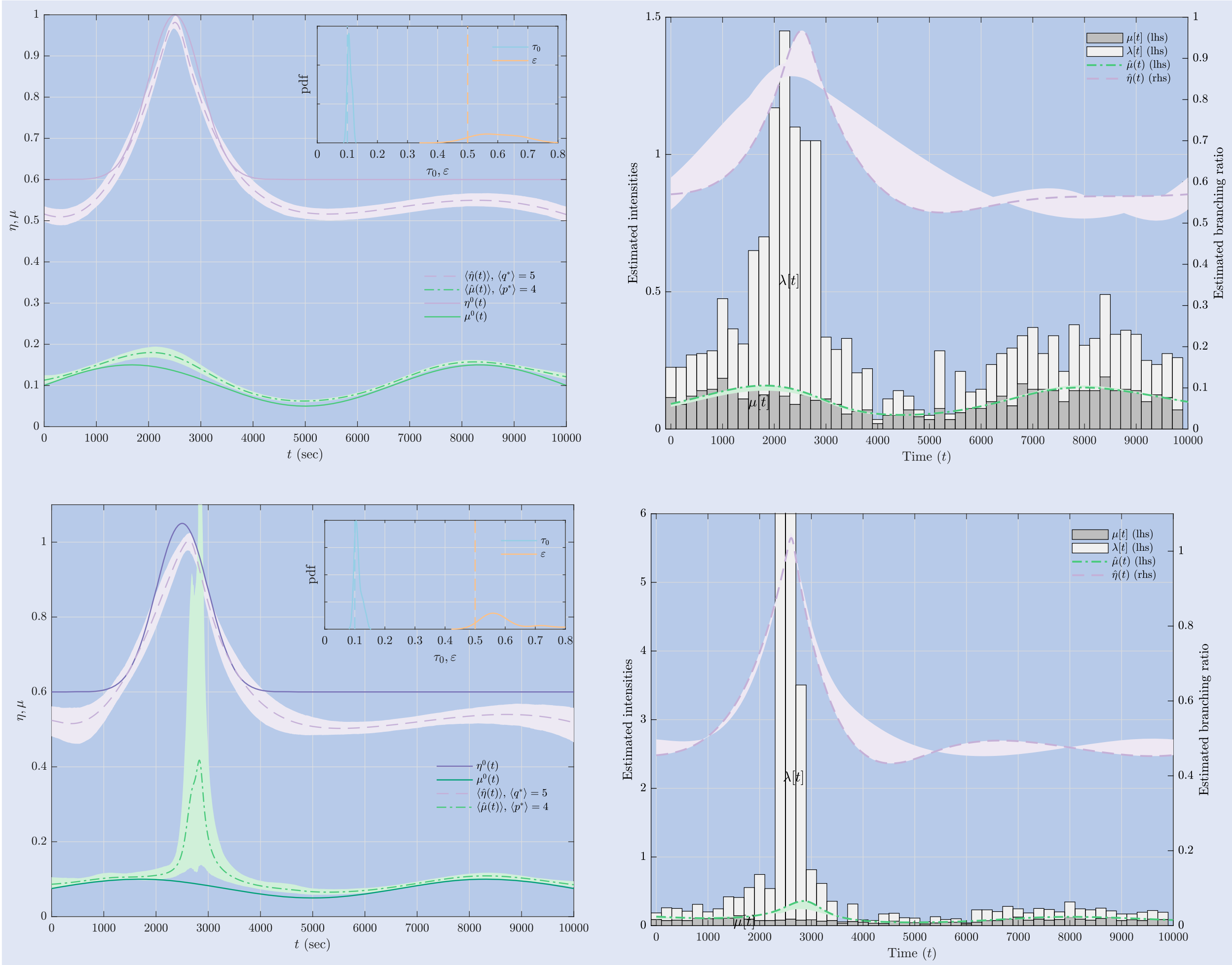

Figure A1. Simulation results for processes with peak branching ratio of $\eta(t) = 1$ (top row) and $\eta(t) = 1.05$ (bottom row). The *left column* shows the true immigration $\mu^0(t)$ and branching ratio $\eta^0(t)$ in solid lines, and their estimates $\hat{\mu}(t)$, $\hat{\eta}(t)$ based on a Hawkes(p, q) obtained from the EM algorithm in dashed lines. For each of the 50 simulations performed, the model with the highest posterior probability (A10) was chosen and the dashed lines represent the average over these estimates. The shaded regions indicate corresponding 95% confidence intervals for the mean, obtained by bootstrapping 1000 times from the resulting function estimates. We also provide the average selected degrees of freedom for the immigration $\langle p^*\rangle$ and branching ratio $\langle q^*\rangle$ in the legend. In the inset, the distributions of the optimal offspring density parameter estimates τ_0 and ε are shown. Their true values are indicated with vertical dashed lines. The *right column*, for a single representative simulation, counts the number of immigrants and overall points, and normalizes them into histogram unconditional intensity estimates, labeled $\mu[t]$ and $\lambda[t]$, respectively. The best fit of the immigration intensity $\hat{\mu}(t)$ is given by the green line. Additionally, on the secondary y-axis, the best estimate of the branching ratio $\hat{\eta}(t)$ is shown. The shaded regions indicate 95% credible intervals obtained from bootstrapping 1000 times from all the fitted functions for this realization, with weights proportional to their posterior probability (A10).

For the estimation of a model like (5), the missing data has to be extended to $(\mathbf{Z}, C)$, where C denotes the immigrant process induced by N^μ, and the complete data log-likelihood thus takes the form

$$Q(\boldsymbol{\theta}|\boldsymbol{\omega}, \hat{\boldsymbol{\theta}}^{(l)}) = \mathbb{E}_{\mathbf{Z},C|\boldsymbol{\omega},\hat{\boldsymbol{\theta}}^{(l)}}[\log L^c(\boldsymbol{\theta}|\boldsymbol{\omega}, \mathbf{Z}, C)], \tag{A11}$$

which can be approximated by averaging over a sample $\boldsymbol{c}^{(k)}$, $k = 1, \ldots, K$ of immigrant realizations from a density $\propto p(\boldsymbol{c}|\boldsymbol{\omega}, \hat{\boldsymbol{\theta}}^{(l)})$ as

$$Q(\boldsymbol{\theta}|\boldsymbol{\omega}, \hat{\boldsymbol{\theta}}^{(l)}) \approx \frac{1}{K}\sum_{k=1}^{K} \mathbb{E}_{\mathbf{Z}|\boldsymbol{\omega},\hat{\boldsymbol{\theta}}^{(l)},\boldsymbol{c}^{(k)}}[\log L^c(\boldsymbol{\theta}|\boldsymbol{\omega}, \mathbf{Z}, \boldsymbol{c}^{(k)})]. \tag{A12}$$

To obtain a conditional MCMC sample from the immigrant process, a Metropolis–Hastings algorithm is employed. Given some state $\boldsymbol{c}^{(k)}$ of the Markov chain, each Metropolis–Hastings iteration starts with randomly proposing the birth or death of an immigrant. In case we choose birth, we uniformly select some $c^* \in \boldsymbol{\omega} \setminus \boldsymbol{c}^{(k)}$. The Metropolis–Hastings birth ratio to decide whether c^* is accepted, given some immigrant vector $\boldsymbol{c}$, is obtained from

$$\begin{aligned} & r_b(\boldsymbol{c}, c^*) \\ & = \frac{N_T - |\boldsymbol{c}|}{|\boldsymbol{c}| + 1}\mu(c^*)e^{-\gamma F(T-c^*)} \\ & \quad \prod_{t_i \in \boldsymbol{\omega}\setminus(\boldsymbol{c}\cup c^*)} \left(1 + \frac{\Theta(t_i - c^*)}{\lambda(t_i|\boldsymbol{c}) - \mu(t_i)}\right) / \left(\lambda(c^*|\boldsymbol{c}) - \mu(c^*)\right). \end{aligned} \tag{A13}$$

and we accept $\boldsymbol{c}^{(k+1)} = \boldsymbol{c}^{(k)} \cup c^*$ with acceptance probability $\min\{r_b(\boldsymbol{c}^{(k)}, c^*), 1\}$. A derivation of this result can be found in Wheatley *et al.* (2018). In case we chose death, $c^* \in \boldsymbol{c}^{(k)}$ is removed from the current state of the Markov chain with acceptance probability $\min\{1/r_b(\boldsymbol{c}^{(k)} \setminus c^*, c^*), 1\}$. The Metropolis–Hastings iterations can then be repeated until a sufficiently long chain is available, and the desired K samples be drawn from this chain.

Then, given some $\boldsymbol{c}^{(k)} = (c_1^k, \ldots, c_{n_\mu^k}^k)$ and due to the fact that the ARMA point process, as well as its generalization in the form of the MA-Hawkes(p, q), maps to a branching process, the inner expectation in (A12) decouples analogously to (A3), using branching probabilities which can be evaluated due to random thinning (Ogata 1981) in a similar fashion like (A4) as

$$\pi_{i,j}^{\Theta,k} = \begin{cases} \frac{\Theta(t_i - c_j^k|\hat{\boldsymbol{\theta}}^{(l)})}{\lambda(t_i|\hat{\boldsymbol{\theta}}^{(l)},\boldsymbol{c}^{(k)}) - \mu(t_i|\hat{\boldsymbol{\theta}}^{(l)})} & \text{for } t_i \in \boldsymbol{\omega} \setminus \boldsymbol{c}^{(k)}, t_j \in \boldsymbol{\omega} \\ 0 & \text{otherwise} \end{cases}$$

$$\pi_{i,j}^{h,k} = \begin{cases} \frac{\eta(t_j|\hat{\boldsymbol{\theta}}^{(l)})h(t_i - t_j|\hat{\boldsymbol{\theta}}^{(l)})}{\lambda(t_i|\hat{\boldsymbol{\theta}}^{(l)},\boldsymbol{c}^{(k)}) - \mu(t_i|\hat{\boldsymbol{\theta}}^{(l)})} & \text{for } t_i \in \boldsymbol{\omega} \setminus \boldsymbol{c}^{(k)}, t_j \in \boldsymbol{\omega} \\ 0 & \text{otherwise.} \end{cases} \tag{A14}$$

In the M-step of the algorithm, the objective (A12) is maximized, which—as for estimation without shot-noise effects—has the structure of decoupled density estimation. First, $\hat{\mu}(t)$ is estimated on the MCMC sample of immigration times $\{\boldsymbol{c}^{(1)}, \ldots, \boldsymbol{c}^{(K)}\}$. For a time-varying estimate, a PDF $m(t)$ can be estimated by a density estimator of choice and scaled up by $\hat{N}_T^{\mu} = K^{-1}\sum_{k=1}^{K} n_{\mu}^k$. As for the Hawkes$(p,q)$ estimation presented in the main text, we resort to adaptive logspline density estimates with p degrees of freedom here. A constant immigration intensity estimate is simply obtained from $\hat{N}_T^{\mu}\ /\ T$. Next, the offspring density $h(\cdot)$ is estimated as in the Hawkes(p,q) case with objective (A5), with the difference that the offspring probabilities used in the objective function need to be averaged over the MCMC samples as

$$\pi_{i,j}^{h} = \frac{1}{K}\sum_{k=1}^{K} \pi_{i,j}^{h,k}. \tag{A15}$$

Estimation of the time-dependent branching ratio is then identical to the Hawkes(p,q) case, only using the above average offspring probabilities in the computation of the $\hat{\eta}_j$ in (A6). Estimation of $\Theta(\cdot)$ finally requires to pool inter-event times and weights over all MCMC samples $\boldsymbol{\pi}^{\Theta} := \{(\pi_{i,j}^{\Theta,k}, t_i - c_j^k)\}_{k=1,\ldots,K}$ and maximize

$$\ell_{\Theta}^{c}(\boldsymbol{\theta}|\boldsymbol{\pi}^{\Theta}) = \frac{1}{K}\sum_{k=1}^{K}\sum_{i=1}^{N_T}\sum_{j=1}^{n_{\mu}^k} \pi_{i,j}^{\Theta,k} \log\left(\Theta(t_i - c_j^k)\right) - \frac{\gamma}{K}\sum_{k=1}^{K}\sum_{j=1}^{n_{\mu}^k} F(T - c_j^k), \tag{A16}$$

where $F(t)$ is the cumulative distribution corresponding to the exogenous offspring density $f(t) = \Theta(t)/\gamma$. The resulting full estimation procedure is sketched in the following box.

1: Start from some initial parameter vector $\boldsymbol{\theta}^{(0)} = (\mu(t), \Theta(t), \eta(t), h(t))$
2: Choose an initial set of immigrants $\boldsymbol{c}^{(0)}$ from $\boldsymbol{\omega}$
3: Define some convergence threshold ϵ
4: $l \leftarrow 0$
5: **do**
6: **E-step:**
7: **Generate Markov chain of immigrants:**
8: $u \leftarrow U(0,1)$
9: **if** $u > 0.5$ **then**
10: Choose some $c^* \in \boldsymbol{\omega} \setminus \boldsymbol{c}^{(k-1)}$
11: Compute the birth-ratio $r_b(\boldsymbol{c}^{(k-1)}, c^*)$ as in (A13)
12: Accept $\boldsymbol{c}^{(k)} \leftarrow \boldsymbol{c}^{(k-1)} \cup c^*$ with probability $\min\{r_b(\boldsymbol{c}^{(k-1)}, c^*), 1\}$
13: **else**
14: Choose some $c^* \in \boldsymbol{c}^{(k-1)}$
15: Compute the death-ratio $1/r_b(\boldsymbol{c}^{(k-1)} \setminus c^*, c^*)$ using (A13)
16: Accept $\boldsymbol{c}^{(k)} \leftarrow \boldsymbol{c}^{(k-1)} \setminus c^*$ with probability $\min\{1/r_b(\boldsymbol{c}^{(k-1)} \setminus c^*, c^*), 1\}$
17: **end if**
18: If no modification was accepted in 12 or 16, $\boldsymbol{c}^{(k)} \leftarrow \boldsymbol{c}^{(k-1)}$
19: $k \leftarrow k+1$ and go to 8, until the chain has the desired length
20:
21: Choose K states of the generated Markov chain as samples for the immigrant process
22: $\hat{N}_T^{\mu} \leftarrow K^{-1}\sum_{k=1}^{K} n_{\mu}^k$
23: Compute the probabilistic branching structures $\pi_{i,j}^{\Theta,(k)}$ and $\pi_{i,j}^{h,(k)}$ from (A14)
24: Compute the offspring probabilities $\pi_{i,j}^{h}$ as in (A15)
25:
26: **M-step:**
27: Estimate the density $m(t)$ on $\{\boldsymbol{c}^{(k)}\}_{k=1,\ldots,K}$
28: $\hat{\mu}(t) \leftarrow \hat{N}_T^{\mu} \cdot m(t)$
29: Estimate the offspring density $\hat{h}(t)$ by maximizing (A5) given the current $\pi_{i,j}^{h}$
30: Estimate $\hat{\eta}(t)$ as in (A9) using $\hat{\eta}_j \leftarrow \sum_{i:t_i>t_j} \pi_{i,j}^{h}/\hat{H}(T - t_j|\hat{\boldsymbol{\theta}}^{(l)})$
31: Estimate $\hat{\Theta}(t)$ by maximizing (A16), given $\boldsymbol{\pi}^{\Theta} \leftarrow \{(\pi_{i,j}^{\Theta,k}, t_i - c_j^k)\}_{k=1,\ldots,K}$
32:
33: $l \leftarrow l+1$
34: $\hat{\boldsymbol{\theta}}^{(l)} \leftarrow (\hat{\mu}(t), \hat{\Theta}(t), \hat{\eta}(t), \hat{h}(t))$
35: **while** $\|\hat{\boldsymbol{\theta}}^{(l)} - \hat{\boldsymbol{\theta}}^{(l-1)}\|_{\infty} > \epsilon$
36: $\hat{\boldsymbol{\theta}}^{(l)}$ is the MCEM estimate for the parameters of (5)